AUTOIMMUNE RATIONALIZATION: A PROTECTIVE-SYSTEMS ANALOGY AND A FORMAL MECHANISM FOR SELF-DEFEATING EPISTEMIC PROTECTION

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Abstract. "I believe only what I can prove" sounds like the safest rule a reasoning agent can adopt. We show that, as an acceptance policy, it can defeat the system it protects without a single false verdict. A protective architecture turns finite evidence into verdicts, verdicts into an operational policy, and the policy into the opening or closing of the system's own capabilities, on which its objective depends. Evidence is modelled as refinement relative to a declared class of admitted behaviours: a record can confirm a commitment, refute it, or leave it undecided, and *coliveness* commitments, such as recurrences and responses, can never be confirmed by any finite record although they may be true. Read with *prove* as finite confirmation, the cardinal policy is *default-reject*: it treats "undecided" as "closed", so it never relies on such commitments and keeps the capabilities they gate closed at every time. Confirming them within a narrower context is no lasting remedy: when the context changes, such confirmations can dissolve, and coliveness can return. In a finite game arena we prove that when those capabilities are counterfactually necessary for the objective, the objective becomes unenforceable. We call such protection *self-defeating* when an intended strategy that relies on the protected commitments only where they hold would have won, and separate it from protective trade-offs and from the correct rejection of false commitments. When all the commitments are coliveness and *sustaining*, true on every course of events that achieves the objective, the trade-off case disappears: the cardinal policy is then self-defeating exactly when the capabilities it closes are necessary support, and for every strategy that would have won. Self-defeating protection is the formal object; *autoimmune rationalization* is our name for its reading in an epistemic agent (a rational criterion defeating its own ends), a bounded analogy with selected features of autoimmunity, not an immunological or psychological claim. In its reflexive case, where the agent's reason is read as sustained by the condition its objective secures, the cardinal policy withdraws reliance from that very condition. We also separate the correctness of a criterion from its authority to decide and from the priority of the goods it protects, show that loss through a dependency between protected properties (a protective inversion) coincides in the game model with necessary support of the protected property, and read narratives of the Christian tradition as interpretive cases. Readings in terms of reason, belief, and drama-triangle roles are interpretive.

Key words and phrases: epistemic protection, self-defeating protection, autoimmune rationalization (analogy), finite confirmation, sustaining commitments, acceptance policies, temporal property classes, coliveness, monitorability, drama triangle, Büchi games.

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1. Introduction

"I believe only what I can prove." As a rule for an agent's reasoning this sounds like the safest policy there is. This paper shows that, read precisely, it can be the opposite: a protective policy that, with every verdict correct, withdraws the agent's reliance from the very properties its objective requires, and with it capabilities that the objective needs. The properties at stake are those an agent rests on without ever being able to confirm them by a finite record: that restoration keeps recurring, that every need is eventually met, that every lock it enters it will eventually leave. Such properties are *coliveness*: no finite record confirms them, although they may be true. A policy that relies only on what has been confirmed therefore never relies on them (Theorem 5.3); a default-reject policy relative to a narrower working context may confirm some of them, but that certainty lasts at most as long as the context does (Proposition 7.13). When the capabilities it consequently keeps closed are necessary for the objective, the protection costs the objective it serves (Theorem 6.5); and when the commitments are *sustaining*, satisfied by every course of events that achieves the objective, the loss leaves no room for a protective trade-off and is self-defeating for every conduct that would have won (Proposition 7.11 and Corollary 7.12). The paper is organized around this cardinal instance of what we call *autoimmune rationalization* (Section 7.5). *Interpretive:* under Assumption 5.4, with reason read as a faculty sustained by the condition its objective secures, as the body sustains the mind, the cardinal policy withdraws reliance from that very condition (the reflexive case of Section 7.5). The obstruction concerns coliveness, not liveness as such: an eventuality $\mathbf{F}\ \gamma$ with γ propositional is liveness and is confirmed as soon as γ is observed, and the policy accepts it then.

A protective mechanism can turn against what it protects. Immunology studies the biological case, autoimmunity, in which a breakdown of tolerance directs immune responses at components of the organism itself [GSFV05]. Philosophy has used the same figure more loosely, from Derrida's political *autoimmunity* [Der03] to Esposito's *immunitas*, a protection that can negate the life it protects [Esp11]. Immune metaphors also describe the mind's defences, from the psychological immune system of affective forecasting [GPW$^+$98] to epistemic immunity [PS23], cognitive immunology [Nor21], and a different "cognitive autoimmunity" concerning the confusion of knowledge with ignorance [AM16]. Cognitive science names the protective function itself: *epistemic vigilance*, the mechanisms by which an agent guards against being misinformed [SCH$^+$10]. That an evidential rule forbidding belief ahead of evidence can block truths that depend on prior belief is James's reply to evidentialism [Jam96]; trust that cannot wait for confirmation [Hol94] and self-fulfilling expectations [Mer48] are related themes. The rule itself is classical: Clifford's evidentialism [Cli79] is the position James answers, and Bartley showed that comprehensive rationalism, which accepts only the positions that can be justified, cannot justify that requirement and so "asserts its own untenability" [Bar62, p. 122]. That self-refutation applies the rule to itself. The self-defeat proved here applies a consistent rule, with correct verdicts, to the commitments on which the agent's objective rests, and what is lost is enforceability. In AI safety, proof-gated designs run only what carries a proof [TO23], and proof-based agents meet the procrastination paradox on goals of the form "eventually" [Yud13, FS14]; there the channel is deduction and the failure is trust in an unsound claim, here the channel is finite observation and no verdict is false.

The paper moves through six steps. *Phenomenon*: a protective function, of which "I believe only what I can prove" is the cardinal example, can, by its own criterion and with

no false verdict, defeat the system it protects. The word *rationalization* refers to this operation of a rational criterion itself, not to motivated, after-the-fact self-justification in the psychological sense. *Analogy*: biological autoimmunity, a protective system that damages the organism it protects, offers a structural and functional comparison (Section 2). *Abstraction*: both sides share a protective architecture that turns signals into evidence, evidence into verdicts, verdicts into a policy, and the policy into the opening or closing of capabilities on which an objective depends. *Formal question*: when does locally correct protection remove a capability that its own objective requires? *Formal instrument*: finite evidence as refinement relative to admitted behaviours (Section 3), temporal property classes for what such evidence can settle (Section 4), the separation of verdict from policy (Section 5), and game-theoretic enforceability (Section 6); together they define *self-defeating protection* and, when the unconfirmable commitments are all the commitments, characterize it (Section 7); for sustaining commitments the characterization reduces to necessary support alone, and the cardinal policy is its instance (Section 7.5). *Interpretation*: in an epistemic agent we call this configuration *autoimmune rationalization* (Section 7.3).

In one sentence: a protective system acts on partial evidence; some commitments cannot be confirmed by any finite amount of such evidence; if an architecture converts continued uncertainty into operational rejection, and makes a capability necessary to its own objective depend on that confirmation, then a locally correct protective mechanism can make the global objective unenforceable. The analogy has one principal asymmetry: biological autoimmunity is a breakdown, whereas the mechanism studied here involves no defect, only a policy doing exactly what it was designed to do (Section 2.4). Sections 8 and 9 return to agents, to drama-triangle roles, and to faith and reason, and Section 10 states the limits.

What is inherited and what is new. Inherited from the literature: the safety–liveness decomposition [AS87], the temporal-property hierarchy [MP87, CMP92, MP95], the correspondence between verifiability with certainty and topological openness [Kel96], three-valued runtime verdicts [BLS11], and Büchi-game determinacy and synthesis [McN93, EJ91, GTW02, PR89, RW87]. Guarding an action by a test on what is known, kept separate from the action itself, is the shape of knowledge-based programs [FHMV97]; that an action whose correctness requires common knowledge, which communication that is not guaranteed cannot create, is never taken is shown for coordinated attack [HM90, Prop. 4, Thm. 5, Cor. 6]; and replacing an exact knowledge test by one that is sound but not necessarily complete is already proposed in [EvdMM98, §4]. The idea of judging finite evidence relative to an assumption about the admitted behaviours comes from assumption-based runtime verification [CTT22, HS20]. Inherited from our earlier work: the four property classes and their intersection algebra [Bra26b], and the prefix-level form of that idea, a finite prefix good relative to a declared admitted class, together with the bookkeeping principle that a finite-prefix statement reaches the infinite behaviour only through that class [Bra26c]. New here, relative to these works: the protective-architecture abstraction, with its biological reading and its formal epistemic instance, in which default-reject, whose single difference from tolerance is that it closes on undecided, converts undecidedness into suppression, and a gate places it on the system's own capabilities; necessary support, a counterfactual edge-removal test on winning regions, used here as a measure of what the objective requires that is independent of the regulator; a restricted theorem that carries the pattern of a guarded action that is never taken, because its guard can never be confirmed, to a separate objective: when the moves gated by unconfirmable commitments are necessary support, a sound default-reject

regulator makes an objective that is enforceable with all moves unenforceable under default-reject availability; a conduct-relative test that separates self-defeating protection from a protective trade-off, characterized when the unconfirmable commitments are all the commitments of the architecture; the sustaining case, in which that test collapses and self-defeat is characterized by necessary support alone, with the finite-confirmation policy, "I believe only what I can prove", as its instance; and the autoimmune reading as a typed analogy. That safety-only restriction can block progress (nonblocking supervisory control [RW87], runtime enforcement [LBW05], shields [BKKW15]), that such a loss of liveness can be avoidable when a protector exists that also preserves liveness (liveness-preserving shields [KAB$^+$17, Def. 9], whose alternative protector still enforces safety against every design, whereas our comparison evaluates a declared intended conduct, itself reliance-safe, under tolerant availability), and that restricted sampling can keep favourable evidence from arriving (the hot-stove effect [DM01], self-confirming equilibrium [FL93]) are known; what is new is the composition stated above.

Scope. The formal results concern acceptance policies, commitments as properties of infinite behaviours, and finite two-player arenas. Readings of these objects as reason, belief systems, or drama-triangle roles form an explicitly labelled interpretive model. The paper makes no empirical claim about human populations; the autoimmune vocabulary names the reading of a formally defined mechanism, not a biological mechanism. Section 10 adds a philosophical and anthropological discussion of the authority of criteria and of the priority of goods, and Section 9.2 reads narratives of the Christian tradition as interpretive cases; neither adds a formal claim.

Evidential conventions. Each non-definitional formal statement carries one of two labels, in its proof header or inline after the claim. **PROVEN**: deductively established, here or in a cited source, for all traces or arenas in scope. **WITNESSED**: verified by the test suite of a reference implementation (not by a proof assistant) on a finite, explicitly identified model; this certifies the stated instance, not a uniform generalization. The paper states no formal conjecture; its empirical hypotheses are collected in Section 10.

2. From Biological Autoimmunity to Protective Architectures

2.1. **The organism.** Consider an organism with a protective subsystem. The subsystem never acts on the world as it is; it acts on signals. Receptors register a finite amount of information about what the organism encounters; the subsystem evaluates that information; it activates or withholds a response; the response changes the organism and its surroundings; and new signals follow. We use four features of the immune system and no more. First, whether a response occurs is not settled by a fixed self/non-self distinction alone [PC06]; the danger model is one influential and debated account of this dependence [Mat94]. Second, tolerance of the organism's own components is actively maintained, among other mechanisms by regulatory T cells [SYNO08]. Third, autoimmunity arises when these tolerance mechanisms fail and responses are directed at the organism's own components [GSFV05]. Fourth, memory is produced by the response: responding T-cell clones give rise to memory cells that respond faster on re-encounter [FYR14]. The abstraction makes no claim about any mechanism or disease.

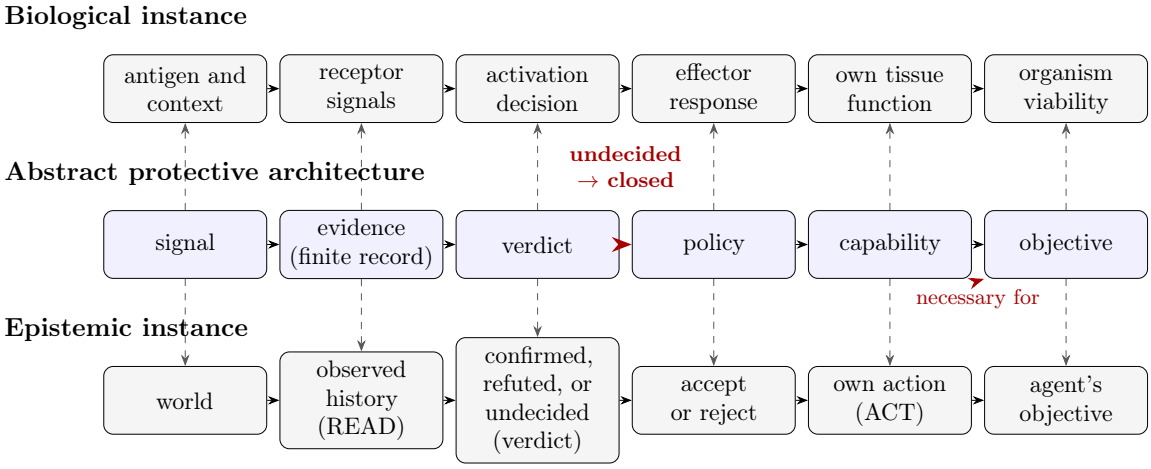


FIGURE 1. The central picture. The abstract protective architecture (middle row), its biological reading (top, informal) and its epistemic instance (bottom, formalized). The biological row is a reading, not a model: immune activation is not a three-valued verdict. The single transition that carries the mechanism of the paper is the policy step turning the verdict "undecided" into the response "closed" (red); it matters when the closed capability is independently necessary for the system's objective. The correspondence is typed row by row: structural in the architectural roles, functional in outcomes and courses; Section 2.4 lists where it fails.

2.2. **The abstract protective architecture.** Between the organism and the agent sits one abstract architecture, and the paper never passes from a cell to a belief without going through it. A protective architecture has six stages: a *signal* from an evolving environment; *evidence*, the finite record accumulated so far; a *verdict* on some question; an operational *policy* that turns verdicts into responses; a *capability* of the system that the response opens or closes; and an *objective* of the system that depends on its capabilities. Figure 1 shows the architecture between its biological reading and its epistemic instance. One transition carries the whole mechanism of this paper: the policy step that turns the verdict "undecided" into the response "closed". When the closed capability is one the system's own objective needs, protection defeats what it protects.

2.3. **The epistemic instance.** In the epistemic instance the host is an agent A. The objects its protective subsystem evaluates are commitments: expectations, trust, and other claims about how things will go, including how the agent itself will go on. Cognitive science studies the protective function as epistemic vigilance [SCH$^+$10]. We group the agent's operations with the five action classes of [BI26, Def. 4.2 and Rem. 6.3]: READ (acquiring evidence; the only class that extends the agent's observed record), THINK (internal computation, which updates the agent's internal state), SPEAK and WRITE (content placed on a transient and on a persistent carrier, readable by other agents or by the agent itself), and ACT (changing the environment). We call MEMORY the internal state that carries information forward; it is state, not an action. Informally, the formal model of this paper

TABLE 1. The mapping H between the biological reading and the epistemic instance. Type: str structural correspondence, the same role (host or stage) in the abstract architecture of Figure 1; fun functional correspondence at the level of outcomes or courses, outside the stages; none no counterpart. In every row the mechanisms differ. No row is an isomorphism: nothing on the biological side is formalized.

Biological instance	**Epistemic instance**	**Type**
organism	agent A	str
immune system	protective subsystem (verdict, policy and gate)	str
receptor signals	finite observed history	str
context-dependent activation	verdict relative to admitted behaviours	str
effector response	policy response: opening or closing a capability	str
the organism's own components	the agent's own capabilities (not "self")	str
impairment of the organism	loss of enforceability of the objective	str
maintained tolerance (antigen-specific, actively maintained)	tolerance of the undecided (policy TOL, a blanket entry; absent under default-reject)	fun
autoimmune disease	self-defeating protection (Definition 7.3), read as autoimmune rationalization	fun
chronic course	persistent self-defeat (Section 7)	fun
immunological memory	persistence of a learned rejection (Section 8; hypothesis only; memory sensitizes, whereas this persistence withholds)	fun
self/non-self discrimination	no counterpart (limit (b))	none

represents READ by the observed history, the verdict as a function of that history, the protective policy as the agent's policy, and ACT by the agent's moves in a game arena; SPEAK and WRITE belong to the general architecture but are not modelled.

The term. We speak of *autoimmune rationalization* when an agent's protective subsystem, acting correctly on finite evidence, suppresses a capability that the agent's own objective needs, in a way that secures nothing the agent's own conduct would not already have secured. The term names a reading, not a formal property: its formal counterpart is self-defeating protection (Definition 7.3), and every result of the paper is stated for that notion. Two routes lead to such a suppression. On the *criterion route*, it follows from the evidential standard and the kind of commitment alone, whatever the history; this route has a formal mechanism (Sections 5 and 6). On the *learned route*, it follows from an internal model shaped by past experience; this route is a modelling hypothesis and is discussed in Section 8.

2.4. **The structural mapping and its limits.** Table 1 states the correspondence between the biological reading and the epistemic instance, each row typed by the kind of correspondence it is.

Where H fails. (a) The epistemic side has no cells, molecules, or aetiology. (b) Modern immunology does not rest on a self/non-self distinction alone [PC06, Mat94], so H maps to the system's own *capabilities*, not to "self". (c) An agent can inspect and revise its evidential standard; an immune system adjusts its activation thresholds only through non-deliberate

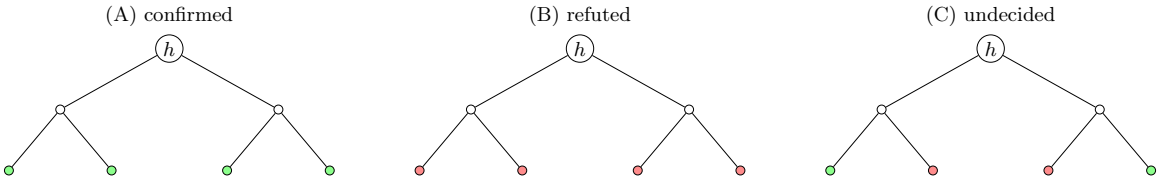


FIGURE 2. Refinement of a finite record h. Branches are admitted continuations, coloured by whether the completed behaviour satisfies the claim (green) or violates it (red). A record confirms a claim when every admitted branch below it is green, refutes it when every admitted branch is red, and leaves it undecided otherwise. A claim no finite record confirms is one for which no node, however deep, has only green branches below it.

regulation. (d) Tissue damage is in principle observable; loss of an objective is a property of the model. (e) The quantitative dynamics of immune responses have no counterpart here. (f) The biological side is a population of cells; the model is a single agent. (g) *Defect versus design.* Biological autoimmunity is usually described as a failure of the mechanisms that maintain tolerance [GSFV05]. The criterial mechanism of this paper involves no defect: a default-reject policy works exactly as designed and has no tolerance of the undecided at all. The mapping therefore concerns an architecture, not a mechanism, and it does not claim that any psychological phenomenon is an autoimmune disease. (h) *Target versus damage.* In organ-specific autoimmunity the recognized antigen typically belongs to the affected tissue, whereas in systemic disease the two can differ; in the model the judged object (a commitment) and the damaged object (a capability) are always distinct, linked only by a declared gate.

3. Finite Evidence as Refinement

The question. A protective subsystem must decide something about a behaviour that is still unfolding. At any moment it holds only a finite record, and each new observation extends the record: $h_0 \sqsubseteq h_1 \sqsubseteq h_2 \sqsubseteq \cdots$, where $\sqsubseteq$ is the prefix order. The question is when the record accumulated so far settles a claim about the potentially unbounded behaviour that is still compatible with it.

A running example. Let the claim be $\xi =$ "the victim lock is eventually left", and let the record so far show the agent inside the lock. Three situations can arise (Figure 2). *(A)* Every behaviour still admitted after the record satisfies ξ: the record *confirms* ξ. *(B)* Every admitted behaviour violates ξ: the record *refutes* ξ. *(C)* Some admitted behaviours satisfy ξ and some violate it: the record leaves ξ *undecided*. For this particular ξ, situation (C) persists after every finite record: however long the agent has stayed in the lock, it may still leave, and it may still stay for ever.

Admitted behaviours. A finite record never determines an arbitrary infinite future by itself. Any inference from finite evidence to a statement about complete behaviours is made relative to the set of continuations the model admits, and changing that set changes what finite evidence can establish. We use the notion of [Bra26c, Def. 3.2]: an *admitted trace class* is a set $T \subseteq \Sigma^\omega$ of infinite traces, regarded as the behaviours the observed system can actually produce, and it must be declared rather than left implicit. Here Σ is the finite alphabet of

observations, one letter per step, and Σ^ω the set of infinite sequences over it. For a finite word u, write $[u] := \{u\sigma : \sigma \in \Sigma^\omega\}$ for its set of infinite extensions. In this paper *refinement* always means extension of the record by further observations; the refinement systems of [Bra26c, Def. 3.1], which act on monitor configurations, are not used.

Definition 3.1 (Evidence relative to admitted behaviours)**.** Let T be an admitted trace class and $\xi \subseteq \Sigma^\omega$ a claim. For a finite record u with $[u] \cap T \neq \emptyset$:

$$u \text{ confirms } \xi \iff [u] \cap T \subseteq \xi, \qquad u \text{ refutes } \xi \iff [u] \cap T \cap \xi = \emptyset,$$

and otherwise u leaves ξ *undecided*. With $T = \Sigma^\omega$ these are the good and bad prefixes of [Bra26b].

In words: a record confirms a claim when no behaviour still admitted after it violates the claim, and refutes it when none satisfies it. The confirming half restates the relative good prefix of [Bra26c, Def. 3.2] in set form. Relative to that paper, two things are added here, both standard in assumption-based runtime verification [CTT22]: the refuting half, a relative bad prefix, and the restriction to records with $[u] \cap T \neq \emptyset$, which excludes records the model does not admit at all, on which both halves would hold vacuously. For the protective architecture this is the point at which the subsystem acts on what remains *possible under its model*, not on the world as it is.

Three sets. For a record u we distinguish $\mathrm{POSSIBLE}(u) = [u]$, all infinite extensions; $\mathrm{ADMITTED}(u) = [u] \cap T$, those the model retains; and, later, $\mathrm{ENFORCEABLE}(q)$, the objectives an agent can guarantee strategically from a position q of an arena (Section 6). The first two are epistemic, the third is strategic, and the paper keeps them apart; the only bridge between them is the coverage requirement of Definition 6.2, which makes every behaviour the arena can produce admitted. Two facts, stated in [Bra26c, Prop. 5.6(i)–(ii), Rem. 5.7] for confirmed predicates along a refinement orbit, govern by analogy what may be inferred here: a confirming record says something about the infinite behaviour only if that behaviour lies in T; and that a condition, once reached, *persists* requires a separate absorption hypothesis. Occurrence is not permanence.

4. What Finite Evidence Can Settle

Temporal property classes answer a single question: what can finite refinement eventually establish about a claim? Each class fixes one side of a claim, its violations or its satisfactions, and says that finite records can always settle that side, or never can. In the four class definitions below $T = \Sigma^\omega$. Claims are written in linear temporal logic: a propositional letter is read at the current step, $\mathbf{G}\,\xi$ means "ξ at every step from now on", $\mathbf{F}\,\xi$ "ξ at some step from now on", and $\mathbf{G}_{\leq k}\,\xi$ "ξ at every step from now through k steps later"; so $\mathbf{G}\,\mathbf{F}\,\mu$ is "μ infinitely often" and $\mathbf{F}\,\mathbf{G}\,\theta$ is "θ from some step on". Four classes answer it [Bra26b, Defs. 3.1–3.4]:

$$\begin{aligned}
\xi \in \mathsf{S} &\iff \forall\sigma \notin \xi\ \exists u \sqsubseteq \sigma\ \forall\tau\,(u \sqsubseteq \tau \Rightarrow \tau \notin \xi),\\
\xi \in \mathsf{C} &\iff \forall\sigma \in \xi\ \exists u \sqsubseteq \sigma\ \forall\tau\,(u \sqsubseteq \tau \Rightarrow \tau \in \xi),\\
\xi \in \mathsf{L} &\iff \forall u\ \exists\tau\,(u \sqsubseteq \tau \wedge \tau \in \xi),\\
\xi \in \mathsf{K} &\iff \forall u\ \exists\tau\,(u \sqsubseteq \tau \wedge \tau \notin \xi).
\end{aligned}$$

TABLE 2. Evidence geometry. Rows: what finite records can do to violations; columns: what they can do to satisfactions. A class is a whole row or column (S = first row, L = last row, C = first column, K = last column); a claim sits in one cell; · marks a cell no claim of this paper occupies. All letters are propositional and non-trivial; α and η are distinct atoms, and α is read at the first position.

	every satisfaction confirmable (C)	some, not all	none confirmable (K)
every violation refutable (S)	$\mathbf{G}_{\leq k}\neg\beta$	$\alpha \vee \mathbf{G}\ \eta$	$\mathbf{G}\ \neg\beta$
some, not all	·	·	·
no violation refutable (L)	$\mathbf{F}\ \gamma$	·	$\mathbf{G}\,\mathbf{F}\ \mu$, $\mathbf{F}\,\mathbf{G}\ \theta$

Read with Definition 3.1: *safety* S says that every violation is eventually refuted by some finite record; *cosafety* C that every satisfaction is eventually confirmed; *liveness* L that no finite record ever refutes the claim; and *coliveness* K that no finite record ever confirms it. The quantifier order matters: S and C are "every trace" statements, so a claim some of whose violations are refutable is not thereby safety [Bra26b, Rem. 3.7]. In the refinement picture of Figure 2, $\xi \in \mathsf{K}$ means that below every node, however deep, some admitted branch violates ξ. Coliveness does not mean that ξ is false: $\mathbf{G}\ \mathbf{F}\ \mu$ may hold on the actual behaviour, and no finite record will show it. In topological terms, K is the class of claims with empty interior, and Theorem 5.3 below is the empty-interior case of the classical correspondence between verifiability with certainty and openness [Kel96].

Classes, not cells. The four classes overlap; the exclusive cells are their intersections [Bra26b]. Table 2 places the claims used in this paper by what finite evidence can do on each side. The entry $\alpha \vee \mathbf{G}\ \eta$ is a regression test for the rest of the paper: it is an unbounded safety claim that is nonetheless confirmed on some behaviours (by an initial α), so "unbounded safety" does not imply "never confirmable". What makes confirmation impossible is K.

Three questions kept apart. Class membership (what finite records can confirm or refute), *monitorability* (whether from every record some extension is decisive [BLS11]), and *acceptance* (what a policy does with a verdict) are different questions. Membership in $\mathsf{L} \cap \mathsf{K}$ is sufficient for non-monitorability, matching the inherently non-monitorable cell of [Bra26b, Rem. 5.8(4)]; it is not necessary, and coliveness alone does not imply it: $\mathbf{G}\ a \in \mathsf{S} \cap \mathsf{K}$ is monitorable, since a prefix containing $\neg a$ refutes it. The mechanism of this paper needs only confirmation, refutation, and undecidedness.

Definition 4.1 (Relative coliveness)**.** For an admitted class T, a claim ξ is *coliveness relative to* T if no record u with $[u] \cap T \neq \emptyset$ confirms ξ in the sense of Definition 3.1.

With $T = \Sigma^\omega$ this is K; the relativised notion is stated here in this form and used from Section 6 on.

The locks. Let v and p mark the victim-lock and persecutor-lock configurations of the drama triangle [Kar68] (Section 9). To be locked is for the lock to hold from some point on: $\mathrm{Lock}_v := \mathbf{F}\ \mathbf{G}\ v$ and $\mathrm{Lock}_p := \mathbf{F}\ \mathbf{G}\ p$.

Proposition 4.2 (Locks lie in liveness ∩ coliveness)**.** $\mathrm{Lock}_v, \mathrm{Lock}_p \in \mathsf{L} \cap \mathsf{K}$*; hence neither is monitorable.*

PROVEN. Every record extends by $(\neg v)^\omega$, which falsifies $\mathbf{F}\,\mathbf{G}\,v$, and by v^ω, which satisfies it. This is the placement [Bra26b, Lemma 5.3] gives to $\mathbf{F}\,\mathbf{G}\,q$ for atomic q, and the same argument applies to any propositional θ with $\theta, \neg\theta$ satisfiable. Non-monitorability follows because no extension of any record is decisive [BLS11]. □

5. From Epistemic Verdict to Operational Policy

Two layers. A verdict is epistemic: confirmed, refuted, or undecided. A response is operational: a capability is opened or closed. The protective architecture needs a rule from the first to the second, and it is at this rule that the mechanism of the paper lives. In the organism, signals are converted into activation decisions; in the agent, verdicts are converted into reliance.

Definition 5.1 (Regulator and policy)**.** A *regulator* is a verdict function $V(u, \xi) \in \{\text{confirmed}, \text{undecided}, \text{refuted}\}$ on finite records u and commitments ξ, relative to a declared admitted class T_R. It is *sound* if $V = \text{confirmed}$ only when u confirms ξ and $V = \text{refuted}$ only when u refutes ξ (Definition 3.1), and *exact* on a set of commitments if it returns exactly the verdict of Definition 3.1 on them. A *protective policy* maps verdicts to $\{\text{open}, \text{closed}\}$, opening on confirmed and closing on refuted. The *default-reject* policy DR also closes on undecided; the *tolerant* policy TOL opens on undecided.

Spelled out:

	confirmed	undecided	refuted
DR	open	**closed**	closed
TOL	open	open	closed

The two policies differ in one entry. Under DR, lack of evidence *for* a commitment is treated operationally like evidence *against* it; in security design this is the classical principle of fail-safe defaults [SS75]. What follows is not a critique of that principle in general, only of its composition with gates on capabilities the system's own objective needs. That conversion, not any lack of knowledge, is what the rest of the paper studies. In the biological reading it corresponds to responding to what has not been shown to be harmless, the opposite of maintained tolerance.

Finite confirmation. The *finite-confirmation* policy Π is DR with the exact regulator and $T_R = \Sigma^\omega$: a commitment is relied upon at time t iff the record so far confirms it. For a commitment Λ write $\mathrm{accept}_\Lambda(t) = 1$ iff the record $u_{0..t}$ observed up to time t confirms Λ.

Lemma 5.2 (Sound regulators and the classes)**.** *Let $T_R = \Sigma^\omega$.*

(1) *A sound regulator confirms ξ only on records on which Π accepts ξ, and an exact regulator confirms on exactly those records.*
(2) *If $\xi \in \mathsf{K}$, no sound regulator ever confirms ξ.*
(3) *Π eventually accepts ξ on every trace that satisfies ξ iff $\xi \in \mathsf{C}$.*
(4) *If $\xi \notin \mathsf{C} \cup \mathsf{K}$, Π eventually accepts ξ on some satisfying traces and never on others.*

PROVEN. (1) and (2) restate soundness and the definition of K. (3) is the definition of C. (4): since $\xi \notin \mathsf{K}$ some record confirms ξ, and every trace extending it is accepted; since $\xi \notin \mathsf{C}$ some satisfying trace has no confirming record. Clause (3) is, in substance, the learning-theoretic fact that a hypothesis is verifiable with certainty exactly when it is open [Kel96]. □

Clauses (3)–(4) concern Π only: a sound regulator may confirm less, and the regulator that is always undecided is sound.

Theorem 5.3 (Permanent non-acceptance)**.** *Let* $\Lambda \in \mathsf{K}$. *Then* $\mathrm{accept}_\Lambda(t) = 0$ *for every* t*, so the acceptance trajectory satisfies* $\mathbf{G}\,\neg\mathrm{accept}_\Lambda$*; under* DR *with any regulator sound relative to* $T_R = \Sigma^\omega$*, every capability gated on* Λ *is closed at every time.*

PROVEN. No finite record confirms Λ (Lemma 5.2(2)), so $\mathrm{accept}_\Lambda(t) = 0$ at every t. The step to $\mathbf{G}\,\neg\mathrm{accept}_\Lambda$ is a statement proved at every time, not an extrapolation from a finite record. Under DR, an unconfirmed commitment is closed. □

Dependency structure. The theorem uses three premises: the class premise $\Lambda \in \mathsf{K}$ [Bra26b, Def. 3.4]; the rule, finite confirmation followed by default-reject; and the observation model, exact observation with every continuation admitted ($T_R = \Sigma^\omega$). The last is essential. *Permanent* means only that $\mathrm{accept}_\Lambda(t) = 0$ at every t on every trace; it follows from the class premise and the fixed rule over a fixed admitted class, and no absorption or persistence assumption about the observed system is involved. Relative to a smaller admitted class a coliveness claim may acquire confirming records, and the theorem does not apply; together with additional structure such as a refinement system, this is how a confirmation that the trace alone cannot supply can be recovered [Bra26c, Def. 3.2, Thm. 5.5]. As mathematics the theorem is close to definitional; its interest lies in what it is a mechanism of, which Sections 6 and 7 make precise.

A reading. Suppose, as a modelling assumption, that the apparatus applying the policy is itself generated and sustained by the commitment it rejects.

Assumption 5.4 (Generative ground, GEN)**.** The apparatus that applies Π is generated and sustained by Λ.

Under Assumption 5.4 the permanent non-acceptance of Theorem 5.3 is directed at the generator of the apparatus that withholds acceptance. This reading is not part of the theorem, no transition system in this paper realizes Assumption 5.4, and the theorem alone does not make an architecture self-defeating: that requires necessary support and a reliance-safe winner (Section 7).

5.1. **Confirmation-gated and refutation-gated protection.** The policy step can be isolated on a single action.

Proposition 5.5 (Confirmation-gated and refutation-gated actions)**.** *Let an action* a *be gated on a satisfiable commitment* ψ*, with* $T_R = \Sigma^\omega$*.*

(1) *Under the confirmation gate (*a *available iff the record confirms* ψ*): if* $\psi \in \mathsf{K}$*,* a *is never available.*

(2) *Under the refutation gate (a available iff the record does not refute ψ): a is available at every time on every trace satisfying ψ; if moreover $\psi \in \mathsf{S}$, then on every violating trace a becomes unavailable after finitely many steps.*
(3) *Exactness: under the confirmation gate, a is unavailable at every time on every trace iff $\psi \in \mathsf{K}$; if $\psi \notin \mathsf{C} \cup \mathsf{K}$ it becomes available on some satisfying traces and never on others; if $\psi \in \mathsf{C}$ it becomes available on every satisfying trace.*

PROVEN. (1) is Lemma 5.2(2). (2): a trace in ψ has no refuting record, and for $\psi \in \mathsf{S}$ every violating trace has one [Bra26b, Def. 3.1]. (3): "if" is (1); "only if" is the definition of K, since after a confirming record a is available; the remaining clauses are Lemma 5.2(3)–(4). □

For $\psi = \mathbf{G}\,\neg\beta$ with propositional, non-trivial β, which lies in $\mathsf{S} \cap \mathsf{K}$ by the argument of [Bra26b, Lemma 5.3], the confirmation gate forbids a for ever, while the refutation gate allows a until ψ is actually violated and detects every violation. For $\alpha \vee \mathbf{G}\,\eta$ the loss under confirmation is only partial. The refutation gate is the falsificationist discipline of [Pop59] applied to action. By complementation, an action *triggered* only by refutation is never triggered when $\psi \in \mathsf{L}$: the dual failure is non-release rather than non-acceptance (**PROVEN**, by complementation of clause (1)). That component-level correctness does not guarantee system-level correctness is a long-standing point in system safety [Lev12]; in supervisory control, nonblockingness must be required separately from safety [RW87]. The proposition is a small instance of that point for epistemic acceptance.

6. Necessary Capability and Global Objective Failure

The question. Theorem 5.3 says that a capability gated on a coliveness commitment stays closed. Whether that matters depends on the system: the closed capability may be one the system never needed. In the organism, an effector response against a component matters only if the organism needs that component. We therefore need an independent measure of what the system's objective requires, and a two-player game supplies it.

6.1. **The arena.** A finite *arena* $\mathcal{G}$ has vertices owned by the *agent* or by the *environment*, an initial vertex q_0, and a labelling λ from vertices to Σ, so that every play induces a trace; every vertex has at least one successor. $\mathrm{Plays}(\mathcal{G})$ is the set of plays from q_0, and $\mathrm{Outcomes}(\sigma)$ the plays consistent with an agent strategy σ. The owner of a vertex chooses the next move. An *objective* Φ is an ω-regular set of plays; Φ is *enforceable* from q if some agent strategy has all its outcomes in Φ, and $R_{\mathrm{win}}(\mathcal{G})$ is the set of such q. For Büchi objectives, which require a set B of vertices to be visited infinitely often, R_{win} is computed by a standard nested fixpoint $\nu Z.\,\mu Y.\,\big((B \cap \mathrm{cpre}(Z)) \cup \mathrm{cpre}(Y)\big)$ [GTW02], where $\mathrm{cpre}(Z)$ is the set of vertices from which the agent can force the next vertex into Z; informally, R_{win} is the largest set from which the agent can always force another visit to B while staying inside the set. It is computable on a finite arena; $\mathcal{G} \setminus E$ removes the agent moves in E, and an agent vertex left without moves is losing.

The reference arena. Six vertices model a regulator's regimes: S_0 moderate, S_1 a suppression trigger, S_2 an over-dominance trigger, V and P the victim and persecutor locks, and X the exit, with moves

$$S_0 \to \{S_0, S_1, S_2\}, \quad S_1 \to \{S_1, V, S_0\}, \quad S_2 \to \{S_2, P, S_0\},$$
$$V \to \{V, X\}, \quad P \to \{P, X\}, \quad X \to \{X, S_0\}.$$

The labels are listed in Appendix A; in particular v holds at V, p at P, and *exit* at X. The environment owns S_0, S_1, S_2. The objective is "every entered lock is eventually left", $\Phi := \mathbf{G}(\mathit{lock} \to \mathbf{F}\ \mathit{exit})$ with $\mathit{lock} := v \vee p$; on this arena it is equivalent to the Büchi objective $\mathbf{G}\,\mathbf{F}\,\neg\mathit{lock}$, because the only non-lock successor of a lock is X. In Arena-Win the agent owns V, P, X; in Arena-Trap the environment owns V and P.

Theorem 6.1 (Enforceability, the lock-trap dichotomy, and monitor–enforce separation)**.**

(1) *The agent can enforce Φ from S_0 iff $S_0 \in R_{\text{win}}(\mathcal{G})$.*
(2) *On any finite arena with a Büchi objective, exactly one holds from a given vertex: the agent enforces Φ, or the environment enforces its complement. On the reference arena this is the lock-trap dichotomy: the agent prevents a permanent lock, or the environment forces one; and R_{win}(Arena-Win) contains all six vertices, and R_{win}(Arena-Trap) $= \{X\}$.*
(3) $\mathbf{F}\,\mathbf{G}$ *lock is not monitorable, yet whether it can be prevented from S_0 is decidable offline by the fixpoint.*

PROVEN; the regions are also computed by a reference implementation. (1) is the definition of R_{win}, and the dichotomy in (2) is Büchi-game determinacy with positional strategies [McN93, EJ91, GTW02]; the controllable and uncontrollable moves are the supervisory-control reading of [RW87]. For the regions: in Arena-Win every lock vertex has an agent-owned move to X, where the agent can stay, so every vertex is winning; in Arena-Trap the environment can reach V through $S_0 \to S_1$ from any S_i and loop there, P is environment-owned and loops as well, and the agent owns only X, so the region is $\{X\}$. The reference game [Bra26d] (12 passing tests) computes the same regions. (3) combines Proposition 4.2 with (1). □

The game separates two questions that the lock seems to fuse: no finite observation settles whether a lock is permanent, but whether it can be prevented is a finite computation.

6.2. **Gates and necessary support.**

Definition 6.2 (Gated architecture)**.** A *gated architecture* consists of an arena $\mathcal{G}$ with objective Φ, a set Com of commitments, a *gate* $g : \text{Com} \to 2^{\text{agent moves}}$, a regulator V with admitted class T_R, and a policy for each commitment (DR unless stated otherwise). The gate is part of the architecture: it says which of the agent's own moves are conditioned on which commitment, and environment moves are never gated. After a history h, a move e is *available* iff it leaves the last vertex of h and every commitment x with $e \in g(x)$ has an open gate at h; plays start at q_0. We require that T_R admits every behaviour the arena can produce, $\lambda(\text{Plays}(\mathcal{G})) \subseteq T_R$ (*coverage*); then every record the arena produces is admitted, so soundness constrains the regulator on every record that matters.

The gate is a policy-level coupling, not a claim that a commitment causes, entails, or justifies a move: it records only that the architecture makes a move available when a commitment is relied upon. It is declared as part of the design under evaluation.

Definition 6.3 (Necessary support)**.** A set E of agent moves is *necessary support* for Φ at q_0 if $q_0 \in R_{\text{win}}(\mathcal{G})$ and $q_0 \notin R_{\text{win}}(\mathcal{G} \setminus E)$.

Necessary support is a property of capabilities, computed from the arena and the objective alone, independently of any regulator. It is counterfactual necessity, not contribution: when two moves substitute for each other, neither is necessary support on its own, while the pair is. By Lemma 6.4 below, every superset of necessary support is again necessary support, so statements "through Y" are most informative for inclusion-minimal Y.

Lemma 6.4 (Monotonicity and decidability)**.** *$R_{\text{win}}(\mathcal{G} \setminus E') \subseteq R_{\text{win}}(\mathcal{G} \setminus E)$ whenever $E \subseteq E'$, and necessary support is decidable by solving two games (two fixpoint computations for Büchi objectives).*

PROVEN; instances also computed by a reference implementation. Every agent strategy in $\mathcal{G} \setminus E'$ is an agent strategy in $\mathcal{G} \setminus E$ with the same outcomes, which gives monotonicity; decidability is decidability of ω-regular games on finite arenas. On Arena-Win, the escape move $V \to X$ alone is necessary support: without it the environment can steer into V, which becomes a trap, and the winning region shrinks to $\{P, X\}$. The pair of escapes $V \to X, P \to X$ is necessary support (region $\{X\}$); the return move $X \to S_0$ is not. Monotonicity is checked over all 64 subsets of the six agent moves, and substitutable escapes are checked on a four-vertex arena (**WITNESSED**), by a solver written separately from the one in [Bra26d] that reproduces its regions [Bra26a]. □

6.3. **From property to objective.** The pieces now compose into one chain, each stage carried by one earlier result.

Theorem 6.5 (Criterial objective loss)**.** *Let $Y \subseteq \text{Com}$ with every $x \in Y$ in K relative to T_R, let $g(Y) := \bigcup_{x \in Y} g(x)$ be necessary support for Φ at q_0, and let the regulator be sound with policy* DR*. Then Φ is not enforceable from q_0 under* DR *availability, although $q_0 \in R_{\text{win}}(\mathcal{G})$. The argument runs through five stages:*

$$\underbrace{x \in \mathsf{K} \text{ rel. } T_R}_{\text{property}} \Rightarrow \underbrace{\text{no record confirms } x}_{\text{evidence}} \Rightarrow \underbrace{\text{DR keeps } g(x) \text{ closed}}_{\text{policy}}$$
$$\Rightarrow \underbrace{\text{moves} \subseteq \mathcal{G} \setminus g(Y)}_{\text{capability}} \Rightarrow \underbrace{q_0 \notin R_{\text{win}}}_{\text{objective}}.$$

PROVEN. The first step is the definition of K relative to T_R (every record admitted by T_R has an admitted extension violating x); the second is soundness; the third is Definition 5.1. For the fourth, other commitments may be rejected after some histories and not others, so the available moves depend on the history, but they are always a subset of the agent moves of $\mathcal{G} \setminus g(Y)$; every strategy under this availability is a strategy there, with a dead end counted as a loss. For the fifth, environment moves are never removed, and since $q_0 \notin R_{\text{win}}(\mathcal{G} \setminus g(Y))$, determinacy gives the environment a strategy that defeats every such agent strategy. □

Only the last step uses the arena, and it uses necessary support, which was computed without reference to the regulator. The global loss is therefore not built into the definition of rejection. With $T_R = \Sigma^\omega$ the hypothesis is ordinary coliveness and the first three stages are Theorem 5.3.

7. Self-Defeating Protection

The question. Theorem 6.5 shows that correct protection can cost an objective. It does not yet show that the cost is a defect. A safety interlock that disables an engine to prevent an explosion also closes a capability the system needs, and it may be exactly right. What separates a protective trade-off from protection that defeats what it serves?

Reliance safety. The protective content of gating a move on a commitment x is that the system should rely on x only when x holds.

Definition 7.1 (Reliance safety)**.** A play is *reliance-safe for* x if, whenever it takes a move of $g(x)$, its trace satisfies x. An agent strategy is *reliance-safe* if all its outcomes are reliance-safe for every commitment in Com.

Proposition 7.2 (Default-reject secures reliance safety)**.** *With a sound regulator and* DR*, every play consistent with* DR *availability is reliance-safe for every commitment.*

PROVEN. A move of $g(x)$ is available only after a record u that confirms x relative to T_R. The play's trace lies in T_R by the coverage requirement and extends u, so it lies in x: the finite-to-infinite step goes through the admitted class, as in [Bra26c, Def. 3.2]; this is the step used in the proof of [Bra26c, Prop. 5.6(i)]. □

Default-reject enforces reliance safety against every strategy. A tolerant policy does not; it leaves reliance safety to the agent's own conduct. The verdict is a function of the record alone, not of the strategy that produced it. The comparison between the two policies is therefore the natural test.

Definition 7.3 (Self-defeating protection)**.** Let the regulator be sound with policy DR, and let σ^* be an agent strategy, the agent's *intended conduct*; "intended" is a label for the declared strategy, and no notion of intention is formalized. The architecture is *self-defeating through* $Y \subseteq \mathrm{Com}$ *for* σ^* if

(SD-a) no agent strategy enforces Φ from q_0 under DR availability; and
(SD-b) σ^* is reliance-safe, is playable when the commitments in Y are treated tolerantly and the rest by DR, and wins Φ.

The comparison changes a single policy entry, undecided→closed into undecided→open, for the commitments in Y; protection against refuted commitments is untouched. For reliance-safe intended conduct, default-reject then secures nothing that the conduct does not already secure, and it costs Φ. Against conduct that is not reliance-safe, the same caution is a genuine trade-off: it enforces reliance safety at the cost of Φ, and whether that is worth it lies outside the model. The protection is self-defeating relative to the holder's objective. In Parfit's classification [Par84] (definitions as quoted in [Ada89, p. 441]) a theory is self-defeating relative to its own aims; default-reject meets its own epistemic aim, since it never relies on a false commitment, and only the reflexive reading of Section 7.5 brings the objective's condition within that aim.

Proposition 7.4 (Characterization)**.** *Let* $\mathrm{Com} = Y$ *and let the regulator be sound.*

(1) *(SD-b) holds iff* σ^* *is reliance-safe and wins* Φ *with all moves available.*
(2) *If every* $x \in Y$ *is in* K *relative to* T_R*, the architecture is self-defeating through* Y *for some* σ^* *iff* $g(Y)$ *is necessary support for* Φ *at* q_0 *and some reliance-safe strategy wins* Φ*.*

PROVEN. (1) Suppose σ^* is reliance-safe and wins with all moves. If σ^* takes a move of $g(x)$ after a history h, some outcome through that move exists, since the arena is total, and it satisfies x; its trace lies in T_R and extends h, so h does not refute x, a sound regulator does not say refuted, and the tolerant gate is open; so σ^* is playable and (SD-b) holds. Conversely, (SD-b) gives a win with all moves, because tolerant availability is a subset of all moves. (2) Under DR a coliveness gate never opens, so DR availability is exactly $\mathcal{G} \setminus g(Y)$ and (SD-a) is $q_0 \notin R_{\text{win}}(\mathcal{G} \setminus g(Y))$; (SD-b) is (1), and it gives $q_0 \in R_{\text{win}}(\mathcal{G})$. □

Relative to conduct. Self-defeat is a statement about an architecture together with a declared intended conduct σ^*, never about the architecture alone. With $T_R = \Sigma^\omega$ every safety expectation $\mathbf{G}\ \neg z$ with z propositional and non-trivial is in K, so, by Proposition 7.4 with $\text{Com} = Y$, a gate on such expectations whose moves are necessary support is self-defeating for every reliance-safe strategy that wins Φ. If the system may adopt conduct that is not reliance-safe, as in instance (b) below, the same default-reject gate is, against that conduct, a genuine protective trade-off, and the model does not say which description is apt. Two consequences of the definitions make the scope explicit (**PROVEN**). Since reliance safety quantifies over every environment reply, once σ^* takes a move of $g(x)$, every outcome of σ^* through that move satisfies x: self-defeat arises only for commitments that the intended strategy itself guarantees, against every environment reply, once it relies on them. And the non-confirmation depends on T_R being wider than the behaviours of σ^*: if σ^* is deterministic and the labelling is injective on histories, then relative to $\lambda(\text{Outcomes}(\sigma^*))$, a class used here only for comparison since coverage excludes it as an admitted class, the record at which σ^* relies on x already confirms x, because every outcome of σ^* through that record takes the move and so satisfies x.

Persistent and transient self-defeat. Whether the closed capability ever becomes usable again is a separate question from whether its closure costs Φ. Self-defeating protection is *persistent* if no move of $g(Y)$ is available at any point of any play consistent with DR, and *transient* otherwise. Coliveness relative to T_R (Definition 4.1) makes it persistent, but persistence and coliveness are separate: in instance (c) below the commitment is cosafety and is confirmed, but only after the moment at which the gated move was needed, and in the transient variant of (c) the gate opens only after Φ is lost. Definition 7.3 does not require persistence. [Bra26c, Rem. 5.7] records the corresponding point for confirmation, that occurrence becomes permanence only under absorption; here, analogously, a gate that opens can open too late.

7.1. **Worked instances.** Each claim below is **PROVEN** by the argument given with it. In each instance $T_R = \Sigma^\omega$, the regulator is exact, and unlisted vertices carry the empty label. The reference solver [Bra26a] (11 passing tests) checks the necessary-support and winning facts of each instance and, in the transient variant, that Φ is lost before the gate opens; the class placements and persistence are argued in the text.

(a) A commitment about one's own conduct. On ARENA-WIN, with $q_0 = S_0$, let $x_V := \mathbf{G}(v \to \mathbf{F}\ \mathit{exit})$, "the victim lock is eventually left", gate the escape $V \to X$. Then $x_V \in \mathsf{L} \cap \mathsf{K}$, the escape is necessary support (Lemma 6.4), and the strategy "at V or P go to X, at X stay" is a reliance-safe winner. By Proposition 7.4 the architecture is self-defeating for that strategy, persistently so by coliveness. If a second commitment $x_{\mathrm{esc}} := \Phi$ also gates $V \to X$ (and $P \to X$), the architecture is self-defeating through the pair $\{x_V, x_{\mathrm{esc}}\}$ but through neither alone: tolerating either alone leaves $V \to X$ closed by the other.

(b) A commitment the agent's own choice can put at risk. Let $q_0 = A$, $A \to \{F, W\}$, $W \to \{W\}$, $F \to \{\mathit{Good}, \mathit{Risky}\}$, $\mathit{Risky} \to \{\mathit{Crash}, \mathit{Good}\}$, $\mathit{Crash} \to \{\mathit{Good}\}$, $\mathit{Good} \to \{A\}$, with the agent owning A, W, F and the environment owning $\mathit{Risky}, \mathit{Crash}, \mathit{Good}$, crash labelled at Crash, and $\Phi = \mathbf{G}\,\mathbf{F}\ \mathit{done}$ with done at Good. Let $x = \mathbf{G}\ \neg \mathit{crash} \in \mathsf{S} \cap \mathsf{K}$ gate $A \to F$, which is necessary support. The strategy that moves $A \to F$ and then goes from F to Good is a reliance-safe winner: relative to it the architecture is self-defeating, persistently so by coliveness. The strategy that goes from F to Risky also wins Φ but may crash while relying on x; relative to it the architecture is not self-defeating, and default-reject is a genuine trade-off.

(c) Late confirmation. Let $E_0 \to A$, $A \to \{\mathit{Go}, \mathit{Fail}\}$, $\mathit{Go} \to C \to \mathit{Ok} \to \mathit{Ok}$, $\mathit{Fail} \to C_2 \to \mathit{Bad} \to \mathit{Bad}$, with the environment owning E_0, c labelled at C and C_2, bad at Bad, and $\Phi = \mathbf{G}\ \neg \mathit{bad}$, $q_0 = E_0$, and the agent owning every other vertex. Let $x = \mathbf{F}\ c \in \mathsf{C} \cap \mathsf{L}$ gate $A \to \mathit{Go}$. At A the verdict is undecided, DR forces Fail, and Φ is lost; going to Go is a reliance-safe winner. The self-defeat is persistent: persistence concerns the availability of gated moves, not confirmation, and although x is confirmed at C_2, no gated move is available there or later. The loss is induced by the admitted class rather than by the arena: every behaviour this arena can produce from A passes a c, so a regulator admitting exactly the arena's behaviours would confirm x at A and open the gate. Replacing the branch $\mathit{Fail} \to C_2 \to \mathit{Bad}$ by $\mathit{Fail} \to \mathit{Bad}_1 \to C_2 \to A$, with bad at Bad_1 and no other change, gives a transient variant: the gate opens at the second visit to A, after Φ is already lost.

7.2. **Two regression tests.**

Sacrifice. Let a shutdown gate mobility on $x = \mathbf{G}\ \neg \mathit{hazard}$, with x its protective aim. If no reliance-safe strategy wins Φ, the architecture is not self-defeating: default-reject buys reliance safety at the cost of Φ, a protective trade-off. The same holds when the interlock exists because declared conduct may not be carried out: conduct that may deviate from σ^* need not be reliance-safe; against deviating conduct that is not reliance-safe the interlock is a trade-off, while for a declared reliance-safe, winning σ^* it remains self-defeating relative to σ^*, and the model does not decide which conduct to expect. If the real aim is weaker, such as "no explosion", the gate checks a stronger commitment than its aim, a mis-specified gate and a different defect.

False commitment. If no play that uses $g(x)$ satisfies x, a reliance-safe strategy never uses $g(x)$. With $\mathrm{Com} = \{x\}$, either no reliance-safe strategy wins, or one wins without $g(x)$ and then wins under default-reject as well; either way the architecture is not self-defeating. That $x \in \mathsf{K}$ does not mean that x is true, good, or should be accepted; the systemic claim arises only through the independently computed dependency of the objective on the gated

capability. In particular, nothing in this paper says that unconfirmable commitments should be accepted.

7.3. **The autoimmune reading.** We use *autoimmune rationalization* for the reading of self-defeating protection in an epistemic agent: a protective architecture, operating correctly by its own evidential criterion, suppresses a capability that the objective it serves requires, where the agent's intended conduct would have kept the protected commitments safe without it. The term names an analogy with selected features of biological autoimmunity. It is not an immunological claim, and it is not a further mathematical property: every formal statement in this paper is about self-defeating protection. The reading has two scales. A single self-defeating configuration, the object of Definition 7.3 and Proposition 7.4, is its static, atomic mechanism. The fully pathological form is iterative: protective closures recur or persist, and the loss of necessary support they cause sustains or strengthens the conditions that call for further protection, a loop that Figure 5 sketches as a model; along such a process the capacity to recover can contract or be lost. A protective trade-off, as in the sacrifice test above, can be a phase from which a trajectory either recovers or, if later closures remove necessary support, escalates; it is not by itself a step of such an escalation. This iterative process is not formalized here. Philosophically, it can be read as the *absolutization of a regulating capacity*: an instrument of discernment acquires an authority that compromises the ends for which it was introduced, as a norm can lose its function when its application is separated from the good it serves (Section 9.2).

Read back through Table 1, the mechanism of Theorem 6.5 and Proposition 7.4 lines up as follows. The protective subsystem is the architecture of verdict, policy and gate; receptor signals are the finite record; the activation decision is the verdict; the effector response is the policy response, with default-reject in the place where maintained tolerance would be, and through the gate it closes the agent's own capabilities; and impairment of the organism is loss of enforceability of Φ. Persistent self-defeat is the closer counterpart of a chronic course, transient self-defeat of a self-limiting episode; this is a reading, not a classification of disease, and a chronic biological course is dynamic, relapsing-remitting or progressive, with self-amplifying damage, whereas persistence here is a static property of a gate.

The correspondence is structural in the architectural roles (host, record, verdict, response, capability, objective), functional in outcomes and courses (disease, chronic course, memory), and never an identity (limits (a)–(h) of Section 2.4). Two differences are informative. Biological autoimmunity is typically described as a failure of tolerance, sometimes initiated by cross-reactive recognition, whereas in the proved mechanism no verdict is false and the policy works as designed (limit (g)); a regulator with unsound verdicts, which rejects what it misrepresents, would be the closer analogue of cross-reactive recognition, and this classificatory route is named but not formalized (Section 10). And the analogy offers no self/non-self boundary: what the regulator cannot tell from the record is whether the agent's conduct is reliance-safe (limit (b)).

7.4. **Dependencies between protected properties.** A protective subsystem can have enough evidence to apply a local rule correctly and still lack an adequate representation of the structural dependencies among the properties it is meant to preserve. This is an insufficiency of *representation*, distinct from the insufficiency of *evidence* that verdicts deal with: necessary support is computed from the arena and the objective, so a subsystem that

does not represent the arena has no basis for computing it. The notions below make one such dependency precise in the arena of Section 6.1. For an objective ψ we write $R^{\psi}_{\mathrm{win}}(\mathcal{G})$ for the set of vertices from which the agent can enforce ψ, so that $R_{\mathrm{win}}(\mathcal{G}) = R^{\Phi}_{\mathrm{win}}(\mathcal{G})$; an agent vertex left without moves is losing, as before.

Definition 7.5 (Structural precondition)**.** Let p_i, p_j be objectives. Property p_j is a *structural precondition* of p_i at q_0, written $p_j \succ_{\mathrm{dep}} p_i$, if every agent strategy that enforces p_i from q_0 in $\mathcal{G}$ also enforces p_j from q_0.

The relation holds trivially when $p_i \subseteq p_j$; it can also hold because of the arena, when every way of guaranteeing p_i that the arena allows passes through guaranteeing p_j.

Definition 7.6 (Protective inversion)**.** Let $q_0 \in R^{p_i}_{\mathrm{win}}(\mathcal{G})$. Closing a set E of agent moves produces a *protective inversion* for p_i if E is necessary support for some p_j with $p_j \succ_{\mathrm{dep}} p_i$, that is, $q_0 \in R^{p_j}_{\mathrm{win}}(\mathcal{G})$ and $q_0 \notin R^{p_j}_{\mathrm{win}}(\mathcal{G} \setminus E)$.

In the intended reading E is closed by a policy that protects p_i, and the protection undermines p_i by removing the support of a precondition; this purpose is a reading and plays no role in the definition.

Lemma 7.7 (Loss through a precondition)**.** *If $p_j \succ_{\mathrm{dep}} p_i$ at q_0 and $q_0 \notin R^{p_j}_{\mathrm{win}}(\mathcal{G} \setminus E)$, then $q_0 \notin R^{p_i}_{\mathrm{win}}(\mathcal{G} \setminus E)$.*

PROVEN. Suppose an agent strategy σ enforces p_i from q_0 in $\mathcal{G} \setminus E$. No outcome of σ reaches an agent vertex left without moves, since such plays are lost; so σ chooses only moves of $\mathcal{G}$, and, environment moves being unchanged, it is a strategy of $\mathcal{G}$ with the same outcomes, as in the proof of Lemma 6.4. Hence σ enforces p_i in $\mathcal{G}$, so p_j by $p_j \succ_{\mathrm{dep}} p_i$, and, its outcomes being the same, p_j in $\mathcal{G} \setminus E$, a contradiction. □

Proposition 7.8 (Inversion is necessary support)**.** *Let $q_0 \in R^{p_i}_{\mathrm{win}}(\mathcal{G})$. Closing E produces a protective inversion for p_i if and only if E is necessary support for p_i.*

PROVEN. If E is necessary support for p_i, take $p_j := p_i$, since $\succ_{\mathrm{dep}}$ is reflexive. Conversely, Lemma 7.7 gives $q_0 \notin R^{p_i}_{\mathrm{win}}(\mathcal{G} \setminus E)$. □

Corollary 7.9. *In a gated architecture with a sound regulator and* DR*, let* $\mathrm{Com} = Y$ *with every $x \in Y$ in K relative to T_R, let $E := g(Y)$, and take the protected property as the objective, $\Phi := p_i$, with $q_0 \in R^{p_i}_{\mathrm{win}}(\mathcal{G})$. If closing E produces a protective inversion for p_i, then Φ is not enforceable from q_0 under* DR *availability, and the architecture is self-defeating through Y for some σ^* if and only if some reliance-safe strategy wins Φ.*

PROVEN. By Proposition 7.8, E is necessary support for Φ; apply Theorem 6.5 and Proposition 7.4(2). □

Three remarks fix the scope. First, by Proposition 7.8 a protective inversion is not a new class: it coincides with necessary support of the protected property. Its value is diagnostic: it names the precondition whose support the protection removes, which is the dependency a subsystem must represent to see the loss coming. Second, an inversion is not sufficient for self-defeat: Corollary 7.9 still requires a reliance-safe winning strategy, and without one the loss of the capability may be an unavoidable and justified sacrifice (the sacrifice test of the regression tests above). Third, taking the protected property as the objective is a choice of reading; in Definition 6.2 the policy protects reliance on commitments, and the objective is a separate datum.

7.5. **The cardinal policy: "I believe only what I can prove".** The phrase that motivates this paper is a policy before it is a belief. Read in the formal model, with *prove* taken as finite confirmation by the observed record, it is the finite-confirmation policy Π of Section 5: default-reject with the exact regulator and $T_R = \Sigma^\omega$, applied as the agent's acceptance criterion for its commitments. Other readings of *prove* name other evidence channels, to which we return at the end of this subsection. The question here is what Π does to the commitments on which the agent itself rests.

Definition 7.10 (Sustaining commitment)**.** In a gated architecture, a commitment x is *sustaining* for Φ if $\lambda(\rho) \in x$ for every play $\rho \in \Phi$.

A sustaining commitment is one without which the objective is achieved on no play; read as an objective, it is a structural precondition of Φ in the trivial sense noted after Definition 7.5. When $\Phi = \{\rho : \lambda(\rho) \in x\}$ for a commitment x, that commitment is sustaining, and so is every commitment it implies. Recurrences $\mathbf{G}\,\mathbf{F}\,h$ with $\neg h$ satisfiable and responses $\mathbf{G}(n \to \mathbf{F}\,h)$ with $n \wedge \neg h$ satisfiable, for propositional n, h, are coliveness over Σ^ω: every record u extends by the violation $u(\neg h)^\omega$, respectively $u(n \wedge \neg h)^\omega$, as in the proof of Proposition 4.2.

Proposition 7.11 (Sustaining commitments leave no trade-off)**.** *Let* $\mathrm{Com} = Y$*, let the regulator be sound with policy* DR*, and let every* $x \in Y$ *be coliveness relative to* T_R *and sustaining for* Φ*.*

(1) *Every agent strategy that wins* Φ *from* q_0 *in* $\mathcal{G}$ *is reliance-safe.*
(2) *The architecture is self-defeating through* Y *for some* σ^* *iff* $g(Y)$ *is necessary support for* Φ *at* q_0*; in that case it is self-defeating through* Y *for every agent strategy that wins* Φ *from* q_0 *in* $\mathcal{G}$*.*

PROVEN. (1) Every outcome ρ of a winning strategy lies in Φ, so $\lambda(\rho) \in x$ for every $x \in Y$ (Definition 7.10), in particular whenever ρ takes a move of $g(x)$ (Definition 7.1). (2) If the architecture is self-defeating for some σ^*, $g(Y)$ is necessary support by Proposition 7.4(2). Conversely, if $g(Y)$ is necessary support then $q_0 \in R_{\mathrm{win}}(\mathcal{G})$ (Definition 6.3), so some agent strategy wins Φ from q_0, and every such strategy is reliance-safe by (1). Proposition 7.4(2) then gives self-defeat. For the last claim, let σ win Φ from q_0 in $\mathcal{G}$: (SD-a) holds because under DR a coliveness gate never opens, so DR availability is $\mathcal{G}\backslash g(Y)$ and $q_0 \notin R_{\mathrm{win}}(\mathcal{G}\backslash g(Y))$; (SD-b) holds for σ by Proposition 7.4(1), since σ is reliance-safe by (1) and wins with all moves. □

The conduct-relative test of Definition 7.3 exists to separate self-defeat from a protective trade-off, and the trade-off case needs a winning conduct that is not reliance-safe. For sustaining commitments there is none: whatever wins makes them true. Default-reject secures reliance safety (Proposition 7.2), but here winning already implies it, so the plays on which that protection makes a difference are plays that lose Φ anyway. Instance (b) of Section 7.1 marks the boundary: $x = \mathbf{G}\,\neg \mathit{crash}$ is not sustaining for $\mathbf{G}\,\mathbf{F}\,\mathit{done}$, since some play through *Crash*, such as $(A\,F\,\mathit{Risky}\,\mathit{Crash}\,\mathit{Good})^\omega$, satisfies Φ; not being sustaining is what leaves room for the trade-off found there, though it does not by itself produce one. In the sacrifice test, likewise, a shutdown can be a genuine trade-off only when its commitment is not implied by the objective.

Corollary 7.12 (The cardinal policy)**.** *Let the agent's acceptance policy be* Π*, let* $\mathrm{Com} = Y$ *consist of commitments that are coliveness and sustaining for* Φ*, and let* $g(Y)$ *be necessary support for* Φ *at* q_0*. Then* Φ *is not enforceable from* q_0 *under* Π*, although* $q_0 \in R_{\mathrm{win}}(\mathcal{G})$*, and*

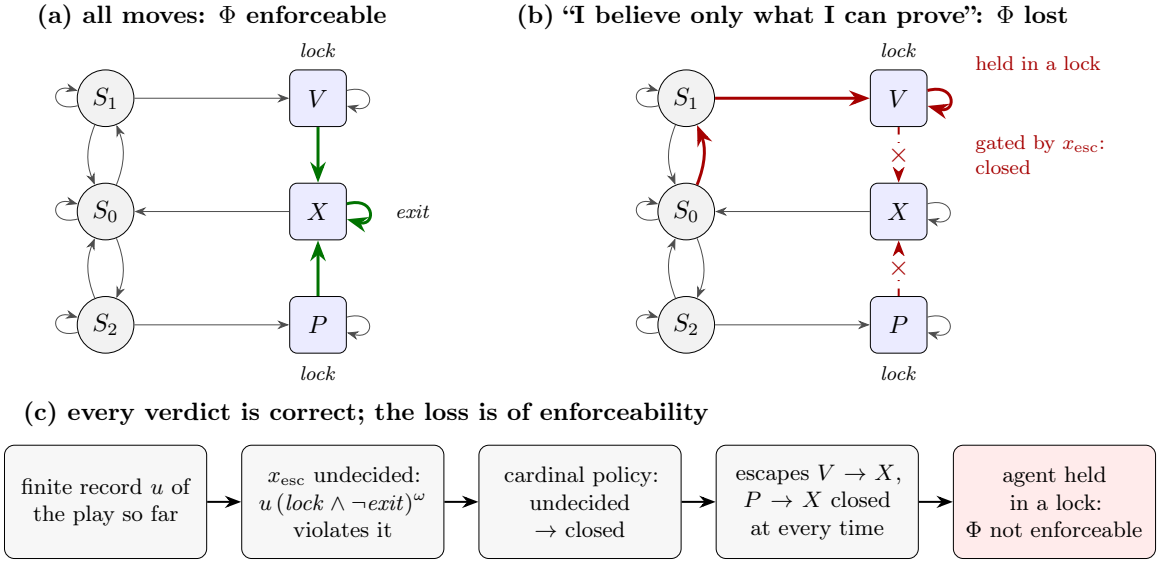


FIGURE 3. The cardinal instance (Section 7.5) on ARENA-WIN. Circles are environment vertices, rounded squares agent vertices; *lock* holds at V and P, *exit* at X, and $\Phi = \mathbf{G}(lock \to \mathbf{F}\ exit)$. (a) With all moves the agent enforces Φ from S_0 by the strategy in green: at V or P go to X, at X stay. (b) The commitment x_{esc}, "every lock I enter I eventually leave", is Φ itself and gates both escapes. Under the finite-confirmation policy it is never confirmed, so the escapes stay closed and the environment can drive the agent into a lock and keep it there (red). (c) The chain, in which no verdict is false. Because x_{esc} is sustaining, every play that wins Φ satisfies it, so the closure secures nothing that a winning conduct does not already secure (Proposition 7.11).

the architecture is self-defeating through Y for every agent strategy that wins Φ from q_0 in $\mathcal{G}$.

PROVEN. The regulator of Π is exact, hence sound; with $T_R = \Sigma^{\omega}$ coverage holds and coliveness relative to T_R is coliveness. Apply Theorem 6.5 and Proposition 7.11(2). □

An instance. On ARENA-WIN with $q_0 = S_0$, let the single commitment be $x_{\text{esc}} := \mathbf{G}(lock \to \mathbf{F}\ exit)$, "every lock I enter I eventually leave", so that $\Phi = \{\rho : \lambda(\rho) \in x_{\text{esc}}\}$, and let it gate both escapes $V \to X$ and $P \to X$. It is coliveness, since $lock \wedge \neg exit$ is satisfiable, and sustaining, since it is the objective's own condition; the pair of escapes is necessary support (Lemma 6.4). By Corollary 7.12 an agent that relies on its own recovery only once that recovery is proven never takes an escape, the environment can hold it in a lock, and the architecture is self-defeating for every strategy that wins in the full arena (Figure 3; **PROVEN**; the necessary support of the pair is argued in Lemma 6.4 and also computed, and the test suite of the reference solver [Bra26a] also checks that the strategy "at V or P go to X, at X stay" wins Φ from S_0 against every environment behaviour, **WITNESSED**). This is instance (a) of Section 7.1 with the commitment x_V removed; the victim lock of Section 9.1 appears here as a consequence of the cardinal policy, not of any false belief.

The reflexive case. *Interpretive.* Take, in Assumption 5.4, the ground to be the objective's own condition: a commitment $\Lambda \in Y$ with $\Phi = \{\rho : \lambda(\rho) \in \Lambda\}$, as x_{esc} in the instance above. Reason, in the canonical analogy, is produced by the body it is meant to serve and continues only while the body's restorative recurrences continue; here those recurrences are Λ. Then $g(Y)$ is necessary support for Λ itself, and Corollary 7.12 reads: the cardinal policy is an *operational defeater* of the condition on which its holder rests. It does not refute Λ, and no verdict is false; it withdraws reliance from Λ at every time, and through the gate it withdraws capabilities without which Λ cannot be enforced. By determinacy, which the proof of Theorem 6.5 already uses, the environment has a strategy that violates Λ against every conduct available under the policy; under Assumption 5.4 these are plays on which the apparatus that applies the policy is not sustained. For sustaining commitments that are weaker than the objective's condition this last step does not follow, since a play can violate Φ without violating them. Proposition 7.11 and Corollary 7.12 do not use Assumption 5.4; the assumption only makes the defeat reflexive, and no transition system in this paper realizes it.

Which proof. The obstruction belongs to one evidence channel. The cardinal policy is pathological here not because it asks for evidence, which is legitimate (Section 9.2), but because it totalizes one admissible criterion, finite observational confirmation, beyond the commitments for which that criterion is adequate, those whose class admits confirming records. Other channels can confirm what this one cannot: relative to a smaller admitted class, or with a refinement structure, coliveness claims can acquire confirming records [Bra26c, Def. 3.2, Thm. 5.5], and knowledge of a loop certifies a recurrence from one observed period (Appendix B). Finite deductive or structural certification is such a channel; a failure of the observational channel is not a failure of all channels, and the paper does not model the others. Within the model, the two ways out of Corollary 7.12 that keep the gate are to change the policy on undecided for the sustaining commitments, the tolerant entry of Definition 5.1, or to change the admitted class; which, if either, an agent should adopt is not decided here.

When the context changes. The verdicts of Definition 3.1 are relative to the admitted class, and an agent's admitted class is its working context: what it takes the world to allow. Within one architecture the class is fixed and covers the arena. Contexts change, however: when the arena changes so that it produces behaviours that the class excludes, coverage requires a larger class, and coliveness can then re-enter through a commitment that had been confirmed.

Proposition 7.13 (Context revision)**.** *Let $T \subseteq T'$ be admitted classes, ξ a claim, and u a record with $[u] \cap T \neq \emptyset$.*

(1) *If u confirms ξ relative to T', it confirms ξ relative to T. In particular, a confirmation relative to Σ^ω survives every change of admitted class under which u remains admitted.*
(2) *The converse fails. Let h be propositional with h and $\neg h$ satisfiable, and $T_{\mathrm{once}} := \{w : w \models \mathbf{G}(h \to \mathbf{G}\, h)\}$, a context in which what is restored stays restored. Every record that contains h and is admitted by T_{once} (for instance $u = a$ with $a \models h$, since $a^\omega \in T_{\mathrm{once}}$) confirms $\mathbf{G}\,\mathbf{F}\, h$ relative to T_{once}, while $\mathbf{G}\,\mathbf{F}\, h$ is coliveness relative to Σ^ω.*
(3) *Let a gated architecture under* DR *with admitted class T be replaced, after a history h_0, by one on an arena $\mathcal{G}'$ whose admitted class $T' \supseteq T$ covers $\mathcal{G}'$, with a regulator sound relative to T' and policy* DR*. At every history extending h_0 whose record does not confirm*

x relative to T', the gate of x is closed, even if an exact regulator for T had opened it at h_0; if x is coliveness relative to T', it is closed at every such history.

PROVEN. (1) $[u] \cap T \subseteq [u] \cap T' \subseteq \xi$, and $[u] \cap T' \supseteq [u] \cap T \neq \emptyset$. (2) A record admitted by T_{once} that contains h has h at every position after its first h, and every w in $[u] \cap T_{\text{once}}$ satisfies $\mathbf{G}\, h$ from that position, hence $\mathbf{G}\,\mathbf{F}\, h$. Relative to Σ^ω, every record u extends by $u(\neg h)^\omega$, which violates $\mathbf{G}\,\mathbf{F}\, h$. (3) A sound regulator returns *confirmed* only on records that confirm x relative to T', and DR opens only on *confirmed* (Definition 5.1); if x is coliveness relative to T', no record confirms it. □

Inside T_{once}, $\mathbf{G}\,\mathbf{F}\, h$ is equivalent to $\mathbf{F}\, h$: the confirmation in (2) is exactly the absorption hypothesis of Section 3, under which occurrence becomes permanence. Non-trivial recurrences $\mathbf{G}\,\mathbf{F}\, h$ and responses $\mathbf{G}(n \to \mathbf{F}\, h)$, with $n \wedge \neg h$ satisfiable, lie in $\mathsf{L} \cap \mathsf{K}$, so any finite confirmation of them is relative to a class smaller than Σ^ω, and by (3) it lasts at most as long as that class is kept; only confirmations relative to Σ^ω, such as that of $\mathbf{F}\ \gamma$ once γ is observed, do not depend on the context. This qualifies the second way out named above: a narrower admitted class lends confirmations that a change of context can withdraw. An agent that applies DR within a narrower working context relies on what that context lets it confirm; after a change of context, (3) withdraws whatever T' does not confirm, and Theorem 6.5 applies with $T_R = T'$ (Corollary 7.12 when $T' = \Sigma^\omega$). *Interpretive:* this is a further reason, for an agent whose context can change, to keep its indefinitely extended commitments in view above all others: their finite certainty is always borrowed from a context.

8. The Agentic Interpretation

The pipeline. Figure 4 places the formal objects inside an agent. READ acquires the finite record; the verdict is a function of the record, and the policy maps verdicts to open or closed capabilities (Definition 5.1); ACT chooses among the capabilities the policy leaves open; MEMORY carries the internal state forward. Part of what the agent later reads is produced by its own earlier outputs: acts change the world it observes, and records it writes can be read back. Such *endogenous evidence* is how a rejection can feed on itself. The formal model represents READ, the verdict and the policy, and ACT; SPEAK, WRITE and MEMORY are part of the general architecture only.

The learned route. Criterial objective loss needs no history. A second route to the same suppression runs through learning, and we state it as a model, not a result (Figure 5, Table 3). A protective objective leads the agent to generate expectations of the safety kind, "nothing bad of kind β_i ever happens", $\mathbf{G}\ \neg\beta_i$. With β_i propositional and non-trivial these lie in $\mathsf{S} \cap \mathsf{K}$, so the agent's own evidence can refute them but never confirm them. Hence, with $T_R = \Sigma^\omega$, any score over these expectations that can increase only when one of them is confirmed never increases, whatever the update rule (**PROVEN**, from [Bra26b, Def. 3.4]). Reading a downward drift of such a score as coming from the class of the expectations rather than from a biased update is an interpretation. That the drift generalizes to other commitments, as in the attributional account of learned helplessness [AST78], is a hypothesis. Once a commitment is rejected, evidence about relying on it stops arriving, and, as a modelling assumption, a rejection maintained under such policy-gated evidence persists; this is known in substance as the hot-stove effect [DM01], self-confirming equilibrium [FL93], incomplete learning [Rot74], self-perception [Bem67], and performative prediction [PZMDH20].

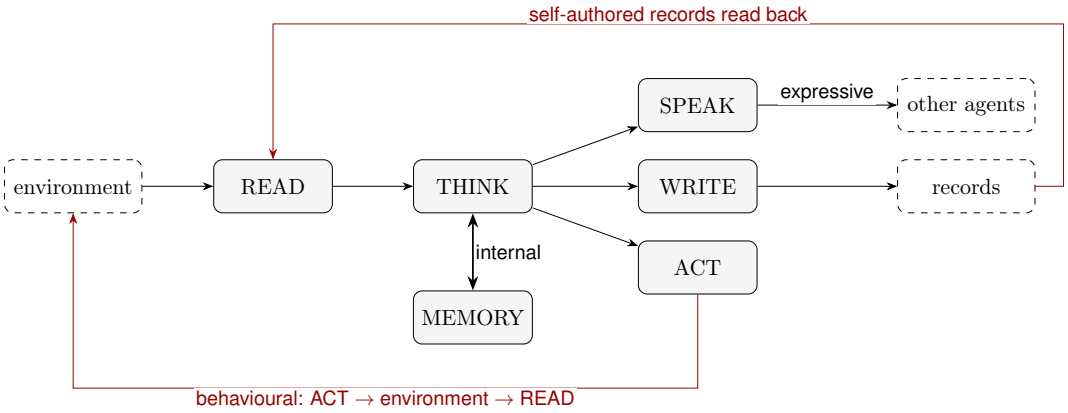


FIGURE 4. The agentic pipeline, a conceptual architecture; action classes after [BI26], and MEMORY is state, not an action. Formalized in this paper: READ (the finite record), the verdict and the policy, ACT (moves in the arena). SPEAK, WRITE and the internal state beyond the record are not modelled. Red arrows are the routes by which the agent's own outputs return as evidence.

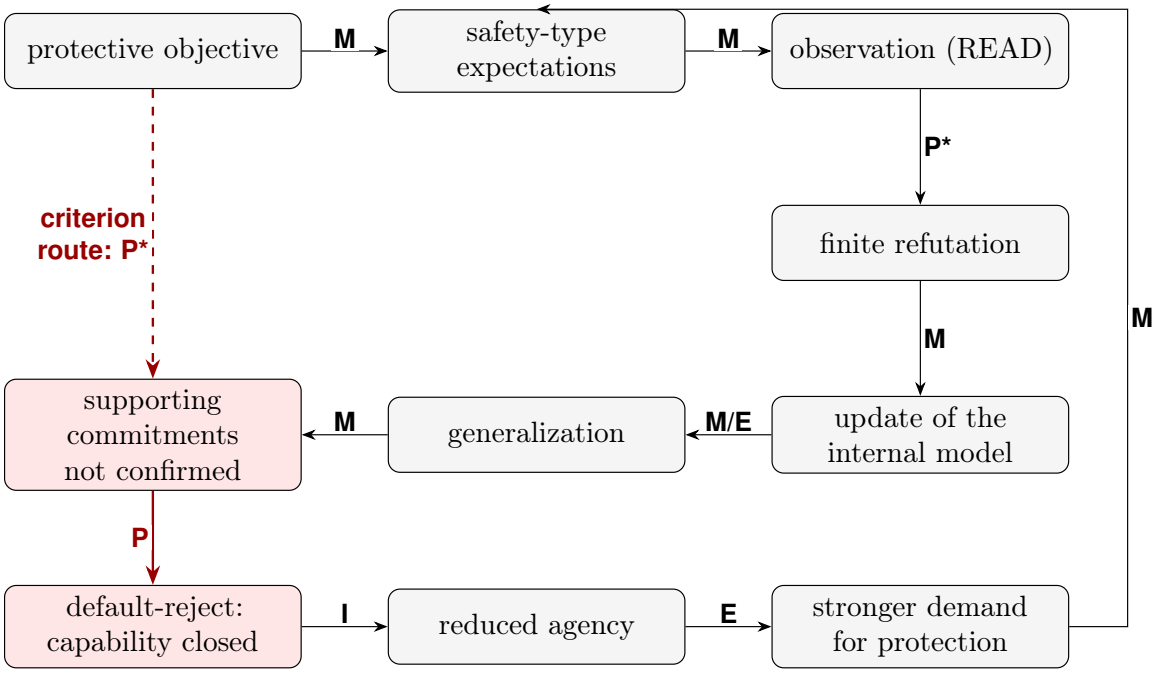


FIGURE 5. A protective suppression loop, a model; it is read as autoimmune rationalization only where (SD-a) and (SD-b) of Definition 7.3 hold. Arrow labels give the epistemic status (Table 3): P proven, P* proven under stated conditions, M modelling assumption, E empirical hypothesis, I interpretive. The dashed arrow is the criterion route of Theorem 5.3. Only the arrows marked P or P* are theorems, the latter under the stated conditions.

TABLE 3. Epistemic status of each arrow of the loop in Figure 5.

Arrow	**Status**	**Basis**
objective → expectations	M; E for agents	modelling choice; hypothesis about agents
observation → finite refutation	P* (definitional)	a violated safety expectation has a refuting record under exact observation
refutation → update	M	the update operator
update → generalization	M; E	motivated by [AST78]; class-induced drift (above)
generalization → not confirmed	M (learned route)	a property of the update
objective → not confirmed	P*, given M (the objective yields coliveness commitments) and M (exact observation, $T_R = \Sigma^\omega$)	Theorem 5.3 (criterion route)
not confirmed → capability closed	P, given the declared gate	default-reject (Definitions 5.1 and 6.2)
closed → reduced agency	I	no agency measure is defined; the formal counterpart is Theorem 6.5
reduced agency → demand	E	no model in this paper

Local correctness, global failure. The theorem is an instance of a general systems pattern. A component that satisfies its local specification—here, never to rely on what is unconfirmed—can, through the way the architecture couples it to other components, make a global objective unenforceable. In supervisory control and system safety the same shape is familiar: local constraints do not guarantee global progress, and nonblockingness must be required separately [RW87, Lev12]. The contribution here is a precise instance: the source of the loss lies neither in the verdicts nor in the objective, but in the composition of a default-reject policy with a gate that makes a necessary capability wait for a confirmation that cannot come.

9. APPLICATIONS

9.1. **Drama-triangle response regimes.** *This subsection is interpretive; the results above do not depend on it.* In the drama triangle [Kar68], a heuristic model from transactional analysis rather than an empirically validated typology, victim, persecutor and rescuer are positions between which an agent moves. Names such as "victim lock", "persecutor lock", and the atoms of Appendix A are mnemonic labels of model states; their interpretive content is supplied by this subsection only. We read each as a response regime of the protective architecture (Table 4); a formalism of roles lies outside this paper. In the self-defeating case, read autoimmunely, two roles meet in one agent: the protective apparatus occupies the persecutor position towards the commitments it rejects, while, on the reference arena, the environment can hold the agent in the victim lock. By contrast, on ARENA-WIN choosing the self-loop $V \to V$ while the escape is open yields a lock by a losing strategy choice in a winnable arena, with no gate involved [Bra26d]. Reference implementations additionally

TABLE 4. Candidate response regimes (interpretive). Signs are relative to the stated target, agent, and time scale.

Regime	**Property classes**	**Main actions**	**Mixed signature**
Victim	demands confirmation; generates $\mathbf{G}\,\neg\beta_i$	THINK (rejection), SPEAK (complaint)	+ short-horizon avoidance; − long-horizon agency; lock $\mathbf{F}\,\mathbf{G}\,v$
Persecutor	imposes safety constraints on another agent	ACT, SPEAK (enforcement)	+ own short-horizon control; − the other's autonomy; lock $\mathbf{F}\,\mathbf{G}\,p$
Rescuer	supplies local relief ($\mathbf{F}$ *relief*)	ACT, SPEAK on the other's behalf	+ the other's short-horizon relief; − the other's autonomy

report a role switch between the two locks under a permutation of the roles, a finite-certificate reading of the recurrent payoff (Racket := $\mathbf{G}\,\mathbf{F}\,r$, which lies in $\mathsf{L} \cap \mathsf{K}$), and an adaptation-based exit on a separate drama-game model [Bra26h, Bra26f, Bra26e]. The objects of those models are defined in the implementations, not in this paper, and the reports carry no formal status here (Appendix B).

9.2. **Faith and reason: a conditional case study.** *Interpretive; no empirical conclusion is drawn.* The question is not whether faith can be proven, but what finite records count as confirming evidence for a particular commitment under the admitted continuations. Only after the commitment is specified can it be classified. "This relationship is and will remain trustworthy", $\mathbf{G}\,\vartheta$ with ϑ propositional and non-trivial, lies in $\mathsf{S} \cap \mathsf{K}$: one betrayal refutes it, no finite record confirms it. "My life will keep proving meaningful", $\mathbf{G}\,\mathbf{F}\,\mu$, lies in $\mathsf{L} \cap \mathsf{K}$. "Things will eventually turn out well", $\mathbf{F}\,\gamma$, lies in $\mathsf{C} \cap \mathsf{L}$ and *is* finitely confirmable. Faith is therefore not a class. *If* a commitment ξ lies in K relative to the admitted class, *if* the moves it gates are necessary support for an objective, and *if* the agent's protective subsystem is sound and default-reject with coverage, then Theorem 6.5 applies, and Definition 7.3 says when the resulting loss is self-defeating. The statement concerns an evidential standard and an architecture, not the correctness of any faith. Theorem 6.5 can be instantiated by a formally specified commitment of several kinds, a working trust, a commitment to a project, or a religious commitment held under a policy that demands finite confirmation, provided every hypothesis holds for that commitment; it says nothing about trust, faith, or belief in general. Definition 7.3 further applies only where the intended strategy guarantees the commitment against every environment reply once it relies on it, and not where the environment alone decides it. The case that fits is the one James described [Jam96]: a commitment whose truth depends on the agent's own reliance on it. That a rule of inference cannot be supplied as one more premise [Car95] suggests, informally, that some acceptance condition must be operative before complete derivation; we record this as interpretation only.

Interpretive cases from the Christian tradition. The tradition offers narratives in which norm, prudence, trust, end and responsibility come into tension. We use them as cases for philosophical analysis, not as empirical evidence and not as instances of the theorems; Table 5 gives, for each, the proposed connection and the limit of the analogy. Two of them deserve a comment. In the sabbath sayings a norm meant to protect a good is read so as to obstruct another fundamental human end; the case separates the validity of a norm, its interpretation,

TABLE 5. Narratives of the Christian tradition as interpretive cases. None is evidence for, or an instance of, a result of the paper.

Case	Proposed connection	Limit of the analogy
The sabbath is made for humankind (Mark 2:27; 3:1–6)	conformity to a norm does not necessarily guarantee that its application preserves the priorities of the normative system as a whole	the narrative does not automatically establish the epistemic correctness of the agents in the technical sense of Definition 5.1
The talents (Matthew 25:14–30)	a policy of risk minimization can prevent the realization of a global objective; local safety differs from generativity and from responsibility in the use of one's capabilities	not, without further assumptions, a case of self-defeat: the risks, the alternatives, and the conditions under which another strategy is preferable must be specified
The good Samaritan (Luke 10:25–37)	acting responsibly under uncertainty does not necessarily require a complete certification of every future consequence	to be kept distinct from legitimate prudence, above all where an unverified intervention could cause harm
Peter on the water (Matthew 14:22–33)	some opportunities for action cannot be fully guaranteed before the action is taken	a theological narrative with a miraculous event; no general justification for taking unverifiable risks
Thomas (John 20:24–29)	a criterion of validation can be appropriate for some classes of propositions without necessarily being a universal procedure; the episode also raises when a criterion can be modified or met by new evidence	not by itself a case of unconfirmability: the confirmation is offered (v. 27)
Francis of Assisi, Canticle of the Creatures	the fundamental properties of an agent need not coincide with what a regulating subsystem can verify; preserving the relations that sustain the agent can be as essential as the local correctness of its decisions	a hymn, not an argument; it bears on the representation of dependencies (Section 7.4)

the scope of its authority, and the hierarchy of ends. Thomas is not to be read as credulous, nor as unreasonable: asking for evidence is legitimate, and the problem arises only when one's own evidence is made the measure of everything. His condition, "except I shall see ... I will not believe" (John 20:25), has the form of the cardinal policy of Section 7.5 restricted to one commitment, and the limit stated in Table 5 applies: there the confirmation is offered.

10. Discussion

Claim level. For sound, default-reject regulators over finite arenas with gated capabilities, the paper proves that correct local protection makes a global objective unenforceable whenever the capabilities gated by unconfirmable commitments are necessary support for it (Theorem 6.5), and, when the unconfirmable commitments are all the commitments, it characterizes when that loss is self-defeating rather than a protective trade-off (Proposition 7.4); when those commitments are sustaining for the objective, no trade-off remains and the finite-confirmation policy, "I believe only what I can prove", is self-defeating for every conduct that would have won (Proposition 7.11 and Corollary 7.12). What makes this more than a restatement of definitions is that necessary support is computed from the arena, independently

of any regulator. The formal ingredients are simple, and the contribution lies in how they are connected. The paper does not prove any of this for agents in general, and it does not prove the loop of Figure 5.

Correctness, authority, and priority. The formal core can be put in one sentence: local epistemic correctness does not guarantee global protective effectiveness. A second sentence is of a different kind: a locally justified criterion is not thereby entitled to govern every decision of the agent. The first is a property of the model, proved under its hypotheses; the second poses a problem of normative justification that needs further assumptions. The decisive distinction is between *the criterion is correct* and *the criterion has the authority to decide this*. It does not lie in the calculus, and it is an ancient one: Sophocles' *Antigone* already stages it, when Antigone denies that Creon's edict has authority over the unwritten laws of the gods (ll. 450–457 [Sop91]). Four notions are worth keeping apart: the *end*, the good the system intends to realize; the *capabilities* through which it is pursued; the *regulating criterion* that authorizes, limits or prevents actions; and the *authority* of the criterion, the set of decisions over which it exercises binding control. Read as an interpretation, they are the objective Φ, the agent's moves, the regulator together with its policy, and the gate, which the model treats as a declared part of the design (Definition 6.2). The problematic configuration arises when the authority of a criterion is extended beyond its domain of justification, to moves that are necessary support for the objective while the criterion can never authorize them, and a reliance-safe conduct would win the objective, as in Definition 7.3. Three levels of evaluation follow.

Level	Central question
Epistemic	Is the verdict correct relative to the available evidence?
Systemic	Does the policy preserve the capabilities the objective needs?
Ethical-anthropological	Which objectives deserve priority, and who is entitled to decide it?

The third level is not automatically reducible to the first two: a system can be epistemically correct and structurally coherent with respect to objectives that are not ethically justified, and an ethically justified end does not by itself make the policy that pursues it adequate. The paper addresses the first two levels, soundness and necessary support with Definition 7.3, and leaves the third outside the model, as it leaves the worth of a protective trade-off. A decision-support system in medicine that blocks a procedure until a complete safety certification is available illustrates the point: the restriction protects against some errors, yet under some conditions delay can compromise other fundamental properties, including the possibility of intervening effectively. This does not mean that the intervention should be authorized automatically. It means that the possible consequences, the operational alternatives and the dependencies among objectives must be represented, which a game model can do, while the cost of delay, the distribution of responsibility and the priorities among objectives lie outside a qualitative model of this kind.

Anticipated objections.

- *"This is the old self-refutation of verificationism."* That argument applies the rule to itself and shows that it cannot justify its own requirement [Bar62, p. 122]. Corollary 7.12 holds for a consistent rule with correct verdicts; what it shows to be lost is not the tenability of the rule but the enforceability of the objective.

- *"The central theorem is trivial."* Its kernel is the empty-interior case of a classical fact [Kel96], and we say so; the sustaining case (Proposition 7.11 and Corollary 7.12) is a direct corollary of the characterization. The content lies in the composition. The test/action separation of knowledge-based programs [FHMV97] and the gate are known; the composition joins them through necessary support, computed independently, and adds the test that separates self-defeat relative to a declared intended strategy from a trade-off.
- *"The gate is arbitrary."* It is a declared part of the design under evaluation; the claim is conditional on it, and the criticism it supports is aimed at architectures that make necessary capabilities wait for impossible confirmation.
- *"Rejecting an unconfirmable commitment is just caution."* Sometimes; the sacrifice test says when, relative to a declared intended strategy. The architecture is self-defeating for an intended conduct only if that conduct is reliance-safe and wins the objective.
- *"The analogy transfers claims from immunology."* It does not (Sections 2.4 and 7.3): the formal results concern self-defeating protection, and autoimmune rationalization is a reading of it; the criterial case is a design without tolerance where biology typically has a defect of tolerance.

Criterial and classificatory failure. Biological autoimmunity is usually described as a failure of tolerance: responses are directed at components that should have been tolerated [GSFV05]. The mechanism proved here is criterial and compositional: no verdict is false, and the loss arises from the policy and the gate. A classificatory route, in which unsound verdicts reject what they misrepresent, and a learned route, in which an internal model shaped by experience withholds confirmation (Section 8), would bring the epistemic case closer to the biological one; both are named, and neither is formalized here. The difference is informative: it shows that self-defeat does not require any faulty component.

Empirical programme. None of the following is tested here.

(1) Agents in a victim-like regime preferentially generate safety-type expectations.
(2) They weight finite refutations more heavily than confirmations.
(3) Commitments that are not finitely confirmable are discounted in proportion to how strongly the agent demands finite confirmation, even without adverse experience.
(4) Interventions that change the acceptance policy, or grant control of an exit action, should outperform interventions that only supply favourable observations (a hypothesis, not a clinical recommendation).
(5) Favourable testimony that does not restrict the admissible continuations should release a learned-route lock but not a criterion-route lock; testing this presupposes an independent assessment of which route an agent is on.

An analogy to misalignment. *Interpretive.* A generator objective that induces a derived controller optimizing a proxy has a similar qualitative shape: if proxy and generator diverge on reachable states, locally optimal updates can violate the generator [AOS$^+$16, Rus19, HvMM$^+$19, SHKK22]. No morphism between the settings is exhibited.

Limitations.

- The results are relative to the declared admitted class $T_R \supseteq \lambda(\mathrm{Plays}(\mathcal{G}))$; a T_R strictly wider than the arena's behaviours can induce a loss that the arena does not force (instance (c) of Section 7.1).

- Self-defeat is relative to the declared arena, admitted class, gate and intended conduct.
- The criterial case is formalized; the learned and classificatory routes are not.
- The mapping H is structural in architectural roles and functional in outcomes, never identical in mechanism, and "reduced agency" is not measured.
- The turn-based game cannot express concurrent de-escalation.
- Witnessed results certify finite reference models, not a uniform class.

11. Conclusion

"I believe only what I can prove", read as the finite-confirmation policy, never relies on a coliveness commitment. A default-reject policy relative to a narrower context may confirm one, but that certainty lasts at most as long as the context: a change of context can dissolve it (Proposition 7.13). When such commitments are sustaining for an objective and the capabilities they gate are necessary support for it, the policy makes the objective unenforceable and is self-defeating for every conduct that would have won (Corollary 7.12); no protective trade-off is left to justify it (Proposition 7.11). This is the cardinal instance of autoimmune rationalization. Read reflexively, under Assumption 5.4 with the objective's own condition as the ground, it is a criterion that operationally defeats the condition on which its holder rests (interpretive; Section 7.5). The general mechanism behind it is as follows.

A protective system acts on finite evidence, and some commitments can never be confirmed by any finite record although they may be true (Section 4). Suppose a sound regulator applies default-reject, and the moves gated by commitments that are coliveness relative to the admitted class are jointly necessary support for an objective. Then the objective, enforceable with all moves, is not enforceable under default-reject availability (Theorem 6.5). Relative to a given intended strategy, this loss is self-defeating when that strategy is reliance-safe, playable with the gated commitments tolerated, and winning (Definition 7.3); when the gated commitments are all the commitments of the architecture, such a strategy exists exactly when their gated moves are necessary support and some reliance-safe strategy wins (Proposition 7.4). Self-defeating protection is the formal object. Read in an epistemic agent, it is the static mechanism of what we call autoimmune rationalization, whose fully pathological form, an iterative process in which such protection sustains the conditions for more of it, is not formalized here (Section 7.3); the reading is a bounded analogy with protective systems that damage the systems they protect, not a further formal property. The theorems are proved in the paper; selected finite instances are additionally checked by reference implementations that are not yet public, and nothing is mechanized in a proof assistant. The readings in terms of reason, belief, and drama-triangle roles remain interpretive, and no empirical, clinical, or biological claim is made. Which objectives deserve priority, and who may decide it, is a further, normative question that the formal results do not settle (Section 10).

12. Future Work

(1) Formalize the classificatory species (unsound verdicts) and the learned route, and test whether a concrete update reproduces the generalization step.
(2) Construct a transition system in which the acceptance apparatus is generated by the ground, so that Assumption 5.4 becomes derived.

(3) Determine the complexity of finding minimal sets of commitments whose tolerance removes self-defeat, and of minimal cuts, and characterize the arenas in which every rejected commitment is necessary support.
(4) Operationalize the empirical programme of Section 10.
(5) Develop role regimes as typed policies, with the drama-triangle regimes as instances.
(6) Formalize the iterative dynamic: a transition system, observables and classification criteria that distinguish proportionate protection, necessary sacrifice, excessive but reversible restriction, and a progressively self-destructive protective circuit, with Definition 7.3 recovered as the single-step case.
(7) Develop a theory of dependencies among capabilities, conditions of existence and objectives (Section 7.4) that keeps structural relations apart from normative judgments about the priority of goods.

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Use of Language Models

This manuscript was prepared with substantial assistance from AI agents (Anthropic Claude models), including drafting, revision and formal checking. The author takes full responsibility for its content.

Data Availability

The reference implementations cited here are the six-state regulator model and its game [Bra26i, Bra26d], the drama-triangle modules [Bra26g, Bra26h, Bra26f, Bra26e], and the necessary-support computation [Bra26a]. The test counts in the text are those of these modules. They are not yet publicly available; an archival release with a persistent identifier is planned, and no identifier has been assigned yet.

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Appendix A. The Reference Model and Bounded Checks

The atoms of the reference arena are f (regulator active), v, p (locks), w (persistent commitment), c (contradiction), *thr* (identity threat), and m (moderate regime). S_0 carries f, m; S_1 carries c; S_2 carries f, w, thr; V and P carry v and f, p; X carries *exit*. The locks are entry-gated (V is entered only from S_1, P only from S_2), absorbing unless left (the only

non-escape moves are self-loops), and have a single escape each. With $\mathbf{F}_{\le k}\chi$ holding at t iff χ holds at some position $t, \dots, t+k$, the bounded onset formulas are

$$\phi_{\text{nihil}} := \mathbf{G}\big((\neg f \wedge c) \to \mathbf{F}_{\le k} v\big), \qquad \psi_{\text{nihil}} := \mathbf{F}\big((\neg f \wedge c) \wedge \mathbf{F}_{\le k} v\big),$$
$$\phi_{\text{dogma}} := \mathbf{G}\big((f \wedge w \wedge \mathit{thr}) \to \mathbf{F}_{\le k} p\big), \qquad \psi_{\text{dogma}} := \mathbf{F}\big((f \wedge w \wedge \mathit{thr}) \wedge \mathbf{F}_{\le k} p\big),$$
$$\phi_{\text{balance}} := \mathbf{G}\big(m \to \neg(\mathbf{F}_{\le k} v \wedge \mathbf{F}_{\le k} p)\big).$$

The ϕ formulas lie in $\mathsf{S} \cap \mathsf{K}$ (every violation has a bad prefix; every prefix extends to a violation by appending a trigger letter without the target and k further such letters, or, for ϕ_{balance}, a letter carrying m, v, p); the ψ formulas lie in $\mathsf{C} \cap \mathsf{L}$ (**PROVEN**). Exhaustive path enumeration at $k = 3$ [Bra26i] (10 passing tests) refutes $\phi_{\text{nihil}}, \phi_{\text{dogma}}$ (a trigger can dwell in its own region), finds ϕ_{balance} satisfied, and confirms $\psi_{\text{nihil}}, \psi_{\text{dogma}}$ with the good prefixes $S_0 S_1 V$ and $S_0 S_2 P$ (**WITNESSED**).

Onset is not permanence. The bounded formulas detect the *onset* of drift towards a lock; the lock $\mathbf{F}\,\mathbf{G}\,v$ is *permanence*. The lassos $L_1 = S_0 S_1 V \cdot V^\omega$ and $L_2 = S_0 S_1 V V V X \cdot X^\omega$ agree through tick 4 and confirm ψ_{nihil} with the same good prefix, yet L_1 satisfies $\mathbf{F}\,\mathbf{G}\,v$ and L_2 does not; on a bare finite prefix the implementation returns UNDECIDED for $\mathbf{F}\,\mathbf{G}\,v$ [Bra26i]. Bounded onset evidence therefore does not determine persistence, as [Bra26c, Prop. 5.6(i)–(ii)] leads us to expect.

Appendix B. Reported Results of the Drama-Game Implementations

The following are reported results of reference implementations whose objects are defined there, not here; they carry no formal status in this paper. A permutation of the three drama roles interchanges the victim-locked and persecutor-locked configurations (12 tests) [Bra26h]. Under the assumption that a drama script is ultimately periodic with a known loop, observing one period certifies the recurrence $\mathbf{G}\,\mathbf{F}\,r$ (11 tests) [Bra26f]; this does not contradict $\mathbf{G}\,\mathbf{F}\,r \in \mathsf{K}$, since the certificate rests on knowledge of the loop, which is an admitted-class assumption in the sense of [Bra26c]. Stuck-lock runs satisfy $\mathbf{F}\,\mathbf{G}(\text{viol})$, and that property is classified in K (19 tests) [Bra26g]. An adaptation rule that raises a readiness parameter by one per round reaches, at a threshold equal to the drama depth, a play that leaves the drama roles recurrently, without editing the observed trace (22 tests) [Bra26e], in contrast to runtime enforcement that truncates [Sch00] or edits [LBW05, FFM12] the execution.

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